11 Vectors

Once again we are completely changing topics from the last chapter. We are going to do a (very) brief introduction to vectors. We'll look at basic notation and concepts involving vectors as well as arithmetic involving vectors. We'll also look at the dot product and cross product of vectors as well as a couple of quick applications of the dot and cross product.

Once we get into the multi-variable Calculus (*i.e.* the topics usually taught in Calculus III) we'll run into vectors on a semi regular basis and so we'll need to be familiar with them and the common notation, concepts and arithmetic involving vectors.

11.1 Basic Concepts

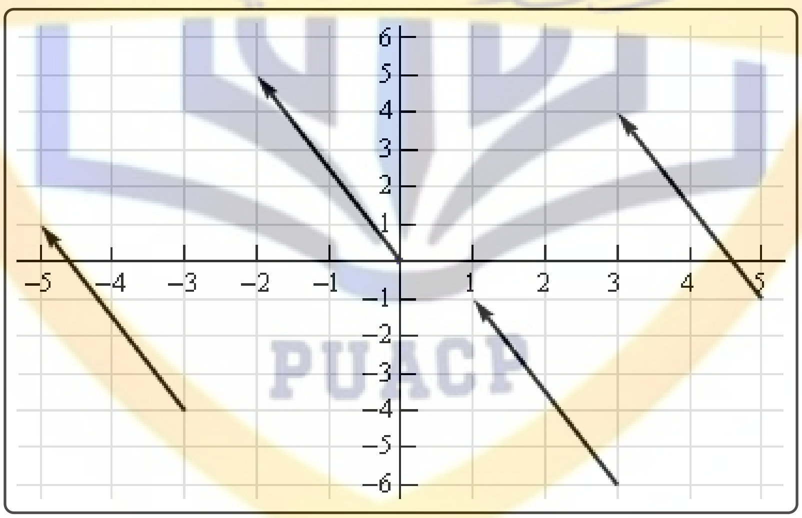
Let's start this section off with a quick discussion on what vectors are used for. Vectors are used to represent quantities that have both a magnitude and a direction. Good examples of quantities that can be represented by vectors are force and velocity. Both of these have a direction and a magnitude.

Let's consider force for a second. A force of say 5 Newtons that is applied in a particular direction can be applied at any point in space. In other words, the point where we apply the force does not change the force itself. Forces are independent of the point of application. To define a force all we need to know is the magnitude of the force and the direction that the force is applied in.

The same idea holds more generally with vectors. Vectors only impart magnitude and direction. They don't impart any information about where the quantity is applied. This is an important idea to always remember in the study of vectors.

In a graphical sense vectors are represented by directed line segments. The length of the line segment is the magnitude of the vector and the direction of the line segment is the direction of the vector. However, because vectors don't impart any information about where the quantity is applied any directed line segment with the same length and direction will represent the same vector.

Consider the sketch below.



Each of the directed line segments in the sketch represents the same vector. In each case the vector starts at a specific point then moves 2 units to the left and 5 units up. The notation that we'll use for this vector is,

$$\vec{v} = \langle -2, 5 \rangle$$

and each of the directed line segments in the sketch are called **representations** of the vector.

Be careful to distinguish vector notation, $\langle -2, 5 \rangle$, from the notation we use to represent coordinates of points, $(-2, 5)$. The vector denotes a magnitude and a direction of a quantity while the point denotes a location in space. So don't mix the notations up!

A representation of the vector $\vec{v} = \langle a_1, a_2 \rangle$ in two dimensional space is any directed line segment, $\overrightarrow{AB}$, from the point $A = (x, y)$ to the point $B = (x + a_1, y + a_2)$. Likewise a representation of the vector $\vec{v} = \langle a_1, a_2, a_3 \rangle$ in three dimensional space is any directed line segment, $\overrightarrow{AB}$, from the point $A = (x, y, z)$ to the point $B = (x + a_1, y + a_2, z + a_3)$.

Note that there is very little difference between the two dimensional and three dimensional formulas above. To get from the three dimensional formula to the two dimensional formula all we did is take out the third component/coordinate. Because of this most of the formulas here are given only in their three dimensional version. If we need them in their two dimensional form we can easily modify the three dimensional form.

There is one representation of a vector that is special in some way. The representation of the vector $\vec{v} = \langle a_1, a_2, a_3 \rangle$ that starts at the point $A = (0, 0, 0)$ and ends at the point $B = (a_1, a_2, a_3)$ is called the **position vector** of the point (a_1, a_2, a_3). So, when we talk about position vectors we are specifying the initial and final point of the vector.

Position vectors are useful if we ever need to represent a point as a vector. As we'll see there are times in which we definitely are going to want to represent points as vectors. In fact, we're going to run into topics that can only be done if we represent points as vectors.

Next, we need to discuss briefly how to generate a vector given the initial and final points of the representation. Given the two points $A = (a_1, a_2, a_3)$ and $B = (b_1, b_2, b_3)$ the vector with the representation $\overrightarrow{AB}$ is,

$$\vec{v} = \langle b_1 - a_1, b_2 - a_2, b_3 - a_3 \rangle$$

Note that we have to be very careful with direction here. The vector above is the vector that starts at A and ends at B. The vector that starts at B and ends at A, *i.e.* with representation $\overrightarrow{BA}$ is,

$$\vec{w} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

These two vectors are different and so we do need to always pay attention to what point is the starting point and what point is the ending point. When determining the vector between two points we always subtract the initial point from the terminal point.

Example 1

Give the vector for each of the following.

(a) The vector from $(2, -7, 0)$ to $(1, -3, -5)$.

(b) The vector from $(1, -3, -5)$ to $(2, -7, 0)$.

(c) The position vector for $(-90, 4)$

Solution

(a) The vector from $(2, -7, 0)$ to $(1, -3, -5)$.

Remember that to construct this vector we subtract coordinates of the starting point from the ending point.

$$\langle 1-2, -3-(-7), -5-0\rangle = \langle -1, 4, -5\rangle$$

(b) The vector from $(1, -3, -5)$ to $(2, -7, 0)$.

Same thing here.

$$\langle 2-1, -7-(-3), 0-(-5)\rangle = \langle 1, -4, 5\rangle$$

Notice that the only difference between the first two is the signs are all opposite. This difference is important as it is this difference that tells us that the two vectors point in opposite directions.

(c) The position vector for $(-90, 4)$

Not much to this one other than acknowledging that the position vector of a point is nothing more than a vector with the point's coordinates as its components.

$$\langle -90, 4\rangle$$

We now need to start discussing some of the basic concepts that we will run into on occasion.

Magnitude

The **magnitude**, or **length**, of the vector $\vec{v} = \langle a_1, a_2, a_3\rangle$ is given by,

$$\|\vec{v}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Example 2

Determine the magnitude of each of the following vectors.

(a) $\vec{a} = \langle 3, -5, 10 \rangle$

(b) $\vec{u} = \left\langle \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right\rangle$

(c) $\vec{w} = \langle 0, 0 \rangle$

(d) $\vec{i} = \langle 1, 0, 0 \rangle$

Solution

There isn't too much to these other than plug into the formula.

(a) $\|\vec{a}\| = \sqrt{9 + 25 + 100} = \sqrt{134}$

(b) $\|\vec{u}\| = \sqrt{\frac{1}{5} + \frac{4}{5}} = \sqrt{1} = 1$

(c) $\|\vec{w}\| = \sqrt{0 + 0} = 0$

(d) $\left\|\vec{i}\right\| = \sqrt{1 + 0 + 0} = 1$

We also have the following fact about the magnitude.

If $\|\vec{a}\| = 0$ then $\vec{a} = \vec{0}$

This should make sense. Because we square all the components the only way we can get zero out of the formula was for the components to be zero in the first place.

Unit Vector

Any vector with magnitude of 1, *i.e.* $\|\vec{u}\| = 1$, is called a **unit vector**.

Example 3

Which of the vectors from Example 2 are unit vectors?

Solution

Both the second and fourth vectors had a length of 1 and so they are the only unit vectors from the first example.

Zero Vector

The vector $\vec{w} = \langle 0, 0 \rangle$ that we saw in the first example is called a zero vector since its components are all zero. Zero vectors are often denoted by $\vec{0}$. Be careful to distinguish 0 (the number) from $\vec{0}$ (the vector). The number 0 denotes the origin in space, while the vector $\vec{0}$ denotes a vector that has no magnitude or direction.

Standard Basis Vectors

The fourth vector from the second example, $\vec{i} = \langle 1, 0, 0 \rangle$, is called a **standard basis vector**. In three dimensional space there are three standard basis vectors,

$$\vec{i} = \langle 1, 0, 0 \rangle \qquad \vec{j} = \langle 0, 1, 0 \rangle \qquad \vec{k} = \langle 0, 0, 1 \rangle$$

In two dimensional space there are two standard basis vectors,

$$\vec{i} = \langle 1, 0 \rangle \qquad \vec{j} = \langle 0, 1 \rangle$$

Note that standard basis vectors are also unit vectors.

Warning

We are pretty much done with this section however, before proceeding to the next section we should point out that vectors are not restricted to two dimensional or three dimensional space. Vectors can exist in general n-dimensional space. The general notation for a n-dimensional vector is,

$$\vec{v} = \langle a_1, a_2, a_3, \ldots, a_n \rangle$$

and each of the a_i's are called **components** of the vector.

Because we will be working almost exclusively with two and three dimensional vectors in this course most of the formulas will be given for the two and/or three dimensional cases. However, most of the concepts/formulas will work with general vectors and the formulas are easily (and naturally) modified for general n-dimensional vectors. Also, because it is easier to visualize things in two dimensions most of the figures related to vectors will be two dimensional figures.

So, we need to be careful to not get too locked into the two or three dimensional cases from our discussions in this chapter. We will be working in these dimensions either because it's easier to visualize the situation or because physical restrictions of the problems will enforce a dimension upon us.

11.2 Vector Arithmetic

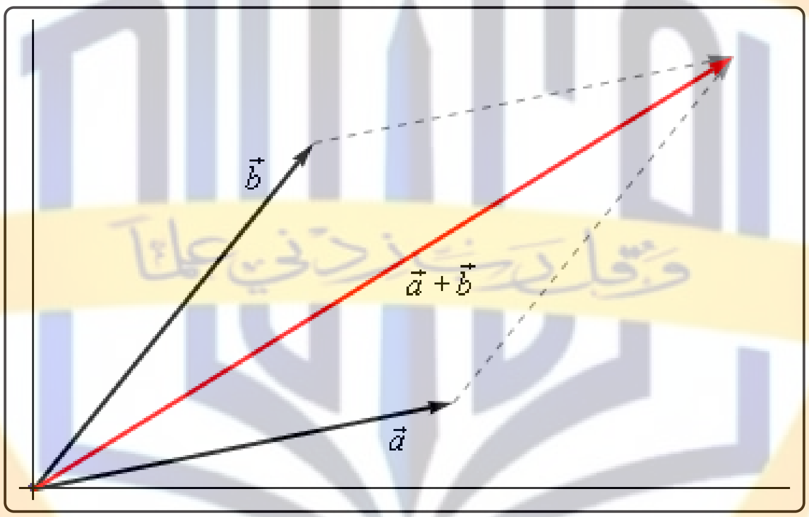
In this section we need to have a brief discussion of vector arithmetic.

We'll start with **addition** of two vectors. So, given the vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ the addition of the two vectors is given by the following formula.

Vector Addition

$$\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

The following figure gives the geometric interpretation of the addition of two vectors.



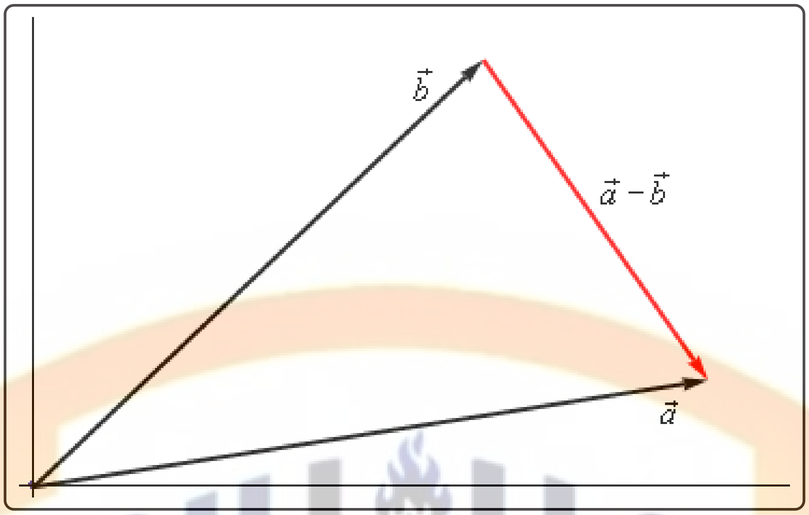
This is sometimes called the **parallelogram law** or **triangle law**.

Computationally, **subtraction** is very similar. Given the vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ the difference of the two vectors is given by,

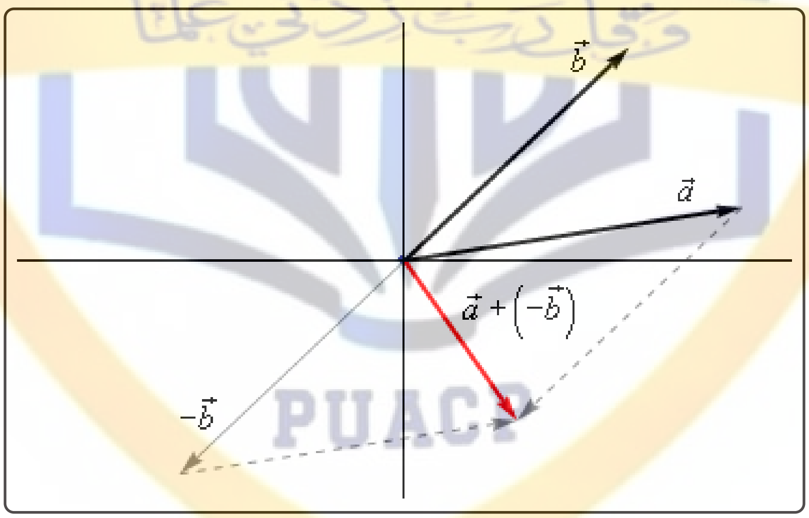
Vector Subtraction

$$\vec{a} - \vec{b} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

Here is the geometric interpretation of the difference of two vectors.



It is a little harder to see this geometric interpretation. To help see this let's instead think of subtraction as the addition of $\vec{a}$ and $-\vec{b}$. First, as we'll see in a bit $-\vec{b}$ is the same vector as $\vec{b}$ with opposite signs on all the components. In other words, $-\vec{b}$ goes in the opposite direction as $\vec{b}$. Here is the vector set up for $\vec{a} + \left(-\vec{b}\right)$.



As we can see from this figure we can move the vector representing $\vec{a} + \left(-\vec{b}\right)$ to the position we've got in the first figure showing the difference of the two vectors.

Note that we can't add or subtract two vectors unless they have the same number of components. If they don't have the same number of components then addition and subtraction can't be done.

The next arithmetic operation that we want to look at is **scalar multiplication**. Given the vector $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and any number c the scalar multiplication is,

Scalar Multiplication

$$c\vec{a} = \langle ca_1, ca_2, ca_3 \rangle$$

So, we multiply all the components by the constant c. To see the geometric interpretation of scalar multiplication let's take a look at an example.

Example 1

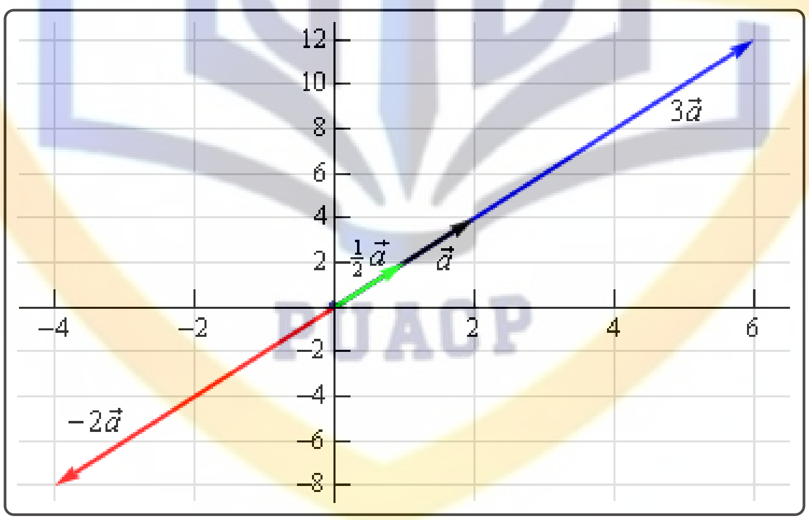
For the vector $\vec{a} = \langle 2, 4 \rangle$ compute $3\vec{a}$, $\frac{1}{2}\vec{a}$ and $-2\vec{a}$. Graph all four vectors on the same axis system.

Solution

Here are the three scalar multiplications.

$$3\vec{a} = \langle 6, 12 \rangle \qquad \frac{1}{2}\vec{a} = \langle 1, 2 \rangle \qquad -2\vec{a} = \langle -4, -8 \rangle$$

Here is the graph for each of these vectors.



In the previous example we can see that if c is positive all scalar multiplication will do is stretch (if $c > 1$) or shrink (if $c < 1$) the original vector, but it won't change the direction. Likewise, if c is negative scalar multiplication will switch the direction so that the vector will point in exactly the opposite direction and it will again stretch or shrink the magnitude of the vector depending upon the size of c.

There are several nice applications of scalar multiplication that we should now take a look at.

The first is parallel vectors. This is a concept that we will see quite a bit over the next couple of sections. Two vectors are parallel if they have the same direction or are in exactly opposite directions. Now, recall again the geometric interpretation of scalar multiplication. When we performed scalar multiplication we generated new vectors that were parallel to the original vectors (and each other for that matter).

Parallel Vectors

So, let's suppose that $\vec{a}$ and $\vec{b}$ are parallel vectors. If they are parallel then there must be a number c so that,

$$\vec{a} = c\vec{b}$$

So, two vectors are parallel if one is a scalar multiple of the other.

Example 2

Determine if the sets of vectors are parallel or not.

(a) $\vec{a} = \langle 2, -4, 1 \rangle$, $\vec{b} = \langle -6, 12, -3 \rangle$

(b) $\vec{a} = \langle 4, 10 \rangle$, $\vec{b} = \langle 2, -9 \rangle$

Solution

(a) $\vec{a} = \langle 2, -4, 1 \rangle$, $\vec{b} = \langle -6, 12, -3 \rangle$

These two vectors are parallel since $\vec{b} = -3\vec{a}$

(b) $\vec{a} = \langle 4, 10 \rangle$, $\vec{b} = \langle 2, -9 \rangle$

These two vectors aren't parallel. This can be seen by noticing that $4\left(\frac{1}{2}\right) = 2$ and yet $10\left(\frac{1}{2}\right) = 5 \neq -9$. In other words, we can't make $\vec{a}$ be a scalar multiple of $\vec{b}$.

The next application is best seen in an example.

Example 3

Find a unit vector that points in the same direction as $\vec{w} = \langle -5, 2, 1\rangle$.

Solution

Okay, what we're asking for is a new parallel vector (points in the same direction) that happens to be a unit vector. We can do this with a scalar multiplication since all scalar multiplication does is change the length of the original vector (along with possibly flipping the direction to the opposite direction).

Here's what we'll do. First let's determine the magnitude of $\vec{w}$.

$$\|\vec{w}\| = \sqrt{25+4+1} = \sqrt{30}$$

Now, let's form the following new vector,

$$\vec{u} = \frac{1}{\|\vec{w}\|}\vec{w} = \frac{1}{\sqrt{30}}\langle -5, 2, 1\rangle = \left\langle -\frac{5}{\sqrt{30}}, \frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}\right\rangle$$

The claim is that this is a unit vector. That's easy enough to check

$$\|\vec{u}\| = \sqrt{\frac{25}{30}+\frac{4}{30}+\frac{1}{30}} = \sqrt{\frac{30}{30}} = 1$$

This vector also points in the same direction as $\vec{w}$ since it is only a scalar multiple of $\vec{w}$ and we used a positive multiple.

So, in general, given a vector $\vec{w}$, $\vec{u} = \frac{\vec{w}}{\|\vec{w}\|}$ will be a unit vector that points in the same direction as $\vec{w}$.

Standard Basis Vectors Revisited

In the previous section we introduced the idea of standard basis vectors without really discussing why they were important. We can now do that. Let's start with the vector

$$\vec{a} = \langle a_1, a_2, a_3\rangle$$

We can use the addition of vectors to break this up as follows,

$$\begin{aligned}\vec{a} &= \langle a_1, a_2, a_3\rangle \\ &= \langle a_1, 0, 0\rangle + \langle 0, a_2, 0\rangle + \langle 0, 0, a_3\rangle\end{aligned}$$

Using scalar multiplication we can further rewrite the vector as,

$$\begin{aligned}\vec{a} &= \langle a_1, 0, 0\rangle + \langle 0, a_2, 0\rangle + \langle 0, 0, a_3\rangle \\ &= a_1\langle 1, 0, 0\rangle + a_2\langle 0, 1, 0\rangle + a_3\langle 0, 0, 1\rangle\end{aligned}$$

Finally, notice that these three new vectors are simply the three standard basis vectors for three dimensional space.

$$\langle a_1, a_2, a_3 \rangle = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

So, we can take any vector and write it in terms of the standard basis vectors. From this point on we will use the two notations interchangeably so make sure that you can deal with both notations.

Example 4

If $\vec{a} = \langle 3, -9, 1 \rangle$ and $\vec{w} = -\vec{i} + 8\vec{k}$ compute $2\vec{a} - 3\vec{w}$.

Solution

In order to do the problem we'll convert to one notation and then perform the indicated operations.

$$\begin{aligned} 2\vec{a} - 3\vec{w} &= 2\langle 3, -9, 1 \rangle - 3\langle -1, 0, 8 \rangle \\ &= \langle 6, -18, 2 \rangle - \langle -3, 0, 24 \rangle \\ &= \langle 9, -18, -22 \rangle \end{aligned}$$

We will leave this section with some basic properties of vector arithmetic.

Properties

If $\vec{v}$, $\vec{w}$ and $\vec{u}$ are vectors (each with the same number of components) and a and b are two numbers then we have the following properties.

$$\vec{v} + \vec{w} = \vec{w} + \vec{v} \qquad \vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$$

$$\vec{v} + \vec{0} = \vec{v} \qquad 1\vec{v} = \vec{v}$$

$$a(\vec{v} + \vec{w}) = a\vec{v} + a\vec{w} \qquad (a + b)\vec{v} = a\vec{v} + b\vec{v}$$

The proofs of these are pretty much just "computation" proofs so we'll prove one of them and leave the others to you to prove.

Proof of $a\left(\vec{v}+\vec{w}\right)=a\vec{v}+a\vec{w}$

We'll start with the two vectors, $\vec{v}=\langle v_1,v_2,\ldots,v_n\rangle$ and $\vec{w}=\langle w_1,w_2,\ldots,w_n\rangle$ and yes we did mean for these to each have n components. The theorem works for general vectors so we may as well do the proof for general vectors.

Now, as noted above this is pretty much just a "computational" proof. What that means is that we'll compute the left side and then do some basic arithmetic on the result to show that we can make the left side look like the right side. Here is the work.

$$\begin{aligned}
a\left(\vec{v}+\vec{w}\right) &= a\left(\langle v_1,v_2,\ldots,v_n\rangle+\langle w_1,w_2,\ldots,w_n\rangle\right)\\
&= a\langle v_1+w_1,v_2+w_2,\ldots,v_n+w_n\rangle\\
&= \langle a\left(v_1+w_1\right),a\left(v_2+w_2\right),\ldots,a\left(v_n+w_n\right)\rangle\\
&= \langle av_1+aw_1,av_2+aw_2,\ldots,av_n+aw_n\rangle\\
&= \langle av_1,av_2,\ldots,av_n\rangle+\langle aw_1,aw_2,\ldots,aw_n\rangle\\
&= a\langle v_1,v_2,\ldots,v_n\rangle+a\langle w_1,w_2,\ldots,w_n\rangle=a\vec{v}+a\vec{w}
\end{aligned}$$

11.3 Dot Product

The next topic for discussion is that of the dot product. Let's jump right into the definition of the dot product. Given the two vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ the dot product is,

Dot Product

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 \tag{11.1}$$

Sometimes the dot product is called the **scalar product**. The dot product is also an example of an inner product and so on occasion you may hear it called an inner product.

Example 1

Compute the dot product for each of the following.

(a) $\vec{v} = 5\vec{i} - 8\vec{j}, \ \vec{w} = \vec{i} + 2\vec{j}$

(b) $\vec{a} = \langle 0, 3, -7 \rangle, \ \vec{b} = \langle 2, 3, 1 \rangle$

Solution

Not much to do with these other than use the formula.

(a) $\vec{v} \cdot \vec{w} = 5 - 16 = -11$

(b) $\vec{a} \cdot \vec{b} = 0 + 9 - 7 = 2$

PUACP

Here are some properties of the dot product.

Properties

$$\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w} \qquad (c\vec{v}) \cdot \vec{w} = \vec{v} \cdot (c\vec{w}) = c\,(\vec{v} \cdot \vec{w})$$

$$\vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v} \qquad \vec{v} \cdot \vec{0} = 0$$

$$\vec{v} \cdot \vec{v} = \|\vec{v}\|^2 \qquad \text{If } \vec{v} \cdot \vec{v} = 0 \text{ then } \vec{v} = \vec{0}$$

The proofs of these properties are mostly "computational" proofs and so we're only going to do a couple of them and leave the rest to you to prove.

Proof of $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$

We'll start with the three vectors, $\vec{u} = \langle u_1, u_2, \ldots, u_n \rangle$, $\vec{v} = \langle v_1, v_2, \ldots, v_n \rangle$ and $\vec{w} = \langle w_1, w_2, \ldots, w_n \rangle$ and yes we did mean for these to each have n components. The theorem works for general vectors so we may as well do the proof for general vectors.

Now, as noted above this is pretty much just a "computational" proof. What that means is that we'll compute the left side and then do some basic arithmetic on the result to show that we can make the left side look like the right side. Here is the work.

$$\begin{aligned}
\vec{u} \cdot (\vec{v} + \vec{w}) &= \langle u_1, u_2, \ldots, u_n \rangle \cdot (\langle v_1, v_2, \ldots, v_n \rangle + \langle w_1, w_2, \ldots, w_n \rangle) \\
&= \langle u_1, u_2, \ldots, u_n \rangle \cdot \langle v_1 + w_1, v_2 + w_2, \ldots, v_n + w_n \rangle \\
&= u_1 (v_1 + w_1) + u_2 (v_2 + w_2) + \ldots + u_n (v_n + w_n) \\
&= u_1 v_1 + u_1 w_1 + u_2 v_2 + u_2 w_2 + \ldots + u_n v_n + u_n w_n \\
&= (u_1 v_1 + u_2 v_2 + \ldots + u_n v_n) + (u_1 w_1 + u_2 w_2 + \ldots + u_n w_n) \\
&= \langle u_1, u_2, \ldots, u_n \rangle \cdot \langle v_1, v_2, \ldots, v_n \rangle + \langle u_1, u_2, \ldots, u_n \rangle \cdot \langle w_1, w_2, \ldots, w_n \rangle \\
&= \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}
\end{aligned}$$

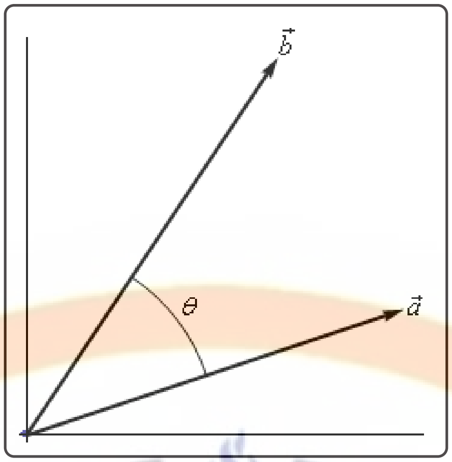
Proof of : If $\vec{v} \cdot \vec{v} = 0$ then $\vec{v} = \vec{0}$

This is a pretty simple proof. Let's start with $\vec{v} = \langle v_1, v_2, \ldots, v_n \rangle$ and compute the dot product.

$$\begin{aligned}
\vec{v} \cdot \vec{v} &= \langle v_1, v_2, \ldots, v_n \rangle \cdot \langle v_1, v_2, \ldots, v_n \rangle \\
&= v_1^2 + v_2^2 + \cdots + v_n^2 \\
&= 0
\end{aligned}$$

Now, since we know $v_i^2 \geq 0$ for all i then the only way for this sum to be zero is to in fact have $v_i^2 = 0$. This in turn however means that we must have $v_i = 0$ and so we must have had $\vec{v} = \vec{0}$.

There is also a nice geometric interpretation to the dot product. First suppose that θ is the angle between $\vec{a}$ and $\vec{b}$ such that $0 \leq \theta \leq \pi$ as shown in the image below.



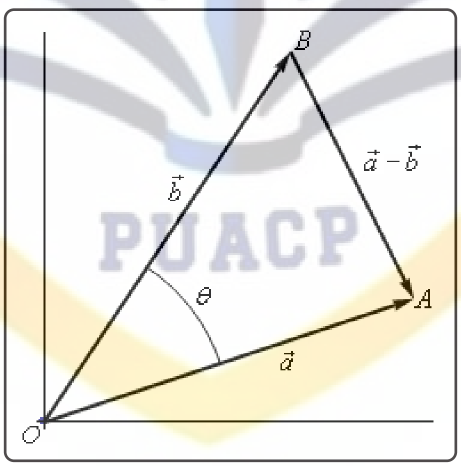
We can then have the following theorem.

Theorem

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \left\|\vec{b}\right\| \cos(\theta) \tag{11.2}$$

Proof

Let's give a modified version of the sketch above.



The three vectors above form the triangle AOB and note that the length of each side is nothing more than the magnitude of the vector forming that side.

The Law of Cosines tells us that,

$$\left\|\vec{a}-\vec{b}\right\|^2 = \|\vec{a}\|^2 + \left\|\vec{b}\right\|^2 - 2\|\vec{a}\| \left\|\vec{b}\right\| \cos(\theta)$$

Also using the properties of dot products we can write the left side as,

$$\begin{aligned}\left\|\vec{a}-\vec{b}\right\|^2 &= \left(\vec{a}-\vec{b}\right)\cdot\left(\vec{a}-\vec{b}\right)\\ &= \vec{a}\cdot\vec{a}-\vec{a}\cdot\vec{b}-\vec{b}\cdot\vec{a}+\vec{b}\cdot\vec{b}\\ &= \|\vec{a}\|^2-2\vec{a}\cdot\vec{b}+\left\|\vec{b}\right\|^2\end{aligned}$$

Our original equation is then,

$$\begin{aligned}\left\|\vec{a}-\vec{b}\right\|^2 &= \|\vec{a}\|^2+\left\|\vec{b}\right\|^2-2\,\|\vec{a}\|\,\left\|\vec{b}\right\|\cos(\theta)\\ \|\vec{a}\|^2-2\vec{a}\cdot\vec{b}+\left\|\vec{b}\right\|^2 &= \|\vec{a}\|^2+\left\|\vec{b}\right\|^2-2\,\|\vec{a}\|\,\left\|\vec{b}\right\|\cos(\theta)\\ -2\vec{a}\cdot\vec{b} &= -2\,\|\vec{a}\|\,\left\|\vec{b}\right\|\cos(\theta)\\ \vec{a}\cdot\vec{b} &= \|\vec{a}\|\,\left\|\vec{b}\right\|\cos(\theta)\end{aligned}$$

The formula from this theorem is often used not to compute a dot product but instead to find the angle between two vectors. Note as well that while the sketch of the two vectors in the proof is for two dimensional vectors the theorem is valid for vectors of any dimension (as long as they have the same dimension of course).

Let's see an example of this.

Example 2

Determine the angle between $\vec{a}=\langle 3,-4,-1\rangle$ and $\vec{b}=\langle 0,5,2\rangle$.

Solution

We will need the dot product as well as the magnitudes of each vector.

$$\vec{a}\cdot\vec{b}=-22 \qquad \|\vec{a}\|=\sqrt{26} \qquad \left\|\vec{b}\right\|=\sqrt{29}$$

The angle is then,

$$\cos(\theta)=\frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\,\left\|\vec{b}\right\|}=\frac{-22}{\sqrt{26}\sqrt{29}}=-0.8011927$$

$$\theta=\cos^{-1}(-0.8011927)=2.5\text{ radians }=143.24\text{ degrees}$$

The dot product gives us a very nice method for determining if two vectors are perpendicular and it will give another method for determining when two vectors are parallel. Note as well that often we will use the term **orthogonal** in place of perpendicular.

Orthogonal/Perpendicular Vectors

Now, if two vectors are orthogonal then we know that the angle between them is 90 degrees. From Equation 11.2 this tells us that if two vectors are orthogonal then,

$$\vec{a} \cdot \vec{b} = 0$$

Likewise, if two vectors are parallel then the angle between them is either 0 degrees (pointing in the same direction) or 180 degrees (pointing in the opposite direction). Once again using Equation 11.2 this would mean that one of the following would have to be true.

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \left\|\vec{b}\right\| \quad (\theta = 0^\circ) \qquad \text{OR} \qquad \vec{a} \cdot \vec{b} = -\|\vec{a}\| \left\|\vec{b}\right\| \quad (\theta = 180^\circ)$$

Example 3

Determine if the following vectors are parallel, orthogonal, or neither.

(a) $\vec{a} = \langle 6, -2, -1 \rangle\,,\ \vec{b} = \langle 2, 5, 2 \rangle$

(b) $\vec{u} = 2\vec{i} - \vec{j},\ \vec{v} = -\frac{1}{2}\vec{i} + \frac{1}{4}\vec{j}$

Solution

(a) $\vec{a} = \langle 6, -2, -1 \rangle\,,\ \vec{b} = \langle 2, 5, 2 \rangle$

First get the dot product to see if they are orthogonal.

$$\vec{a} \cdot \vec{b} = 12 - 10 - 2 = 0$$

The two vectors are orthogonal.

(b) $\vec{u} = 2\vec{i} - \vec{j},\ \vec{v} = -\frac{1}{2}\vec{i} + \frac{1}{4}\vec{j}$

Again, let's get the dot product first.

$$\vec{u} \cdot \vec{v} = -1 - \frac{1}{4} = -\frac{5}{4}$$

So, they aren't orthogonal. Let's get the magnitudes and see if they are parallel.

$$\|\vec{u}\| = \sqrt{5} \qquad \|\vec{v}\| = \sqrt{\frac{5}{16}} = \frac{\sqrt{5}}{4}$$

Now, notice that,

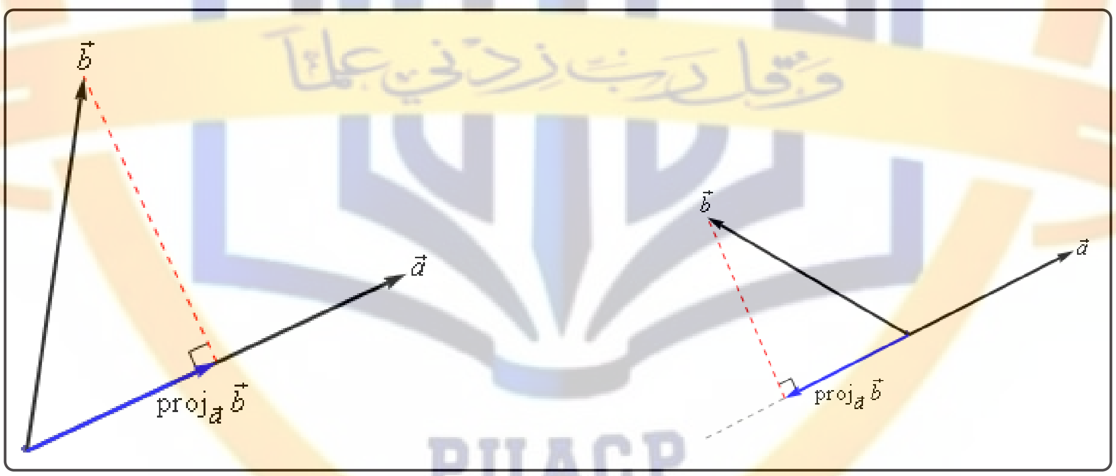
$$\vec{u} \cdot \vec{v} = -\frac{5}{4} = -\sqrt{5}\left(\frac{\sqrt{5}}{4}\right) = -\|\vec{u}\|\ \|\vec{v}\|$$

So, the two vectors are parallel.

There are several nice applications of the dot product as well that we should look at.

Projections

The best way to understand projections is to see a couple of sketches. So, given two vectors $\vec{a}$ and $\vec{b}$ we want to determine the projection of $\vec{b}$ onto $\vec{a}$. The projection is denoted by $\text{proj}_{\vec{a}}\vec{b}$. Here are a couple of sketches illustrating the projection.



So, to get the projection of $\vec{b}$ onto $\vec{a}$ we drop straight down from the end of $\vec{b}$ until we hit (and form a right angle) with the line that is parallel to $\vec{a}$. The projection is then the vector that is parallel to $\vec{a}$, starts at the same point both of the original vectors started at and ends where the dashed line hits the line parallel to $\vec{a}$.

There is a nice formula for finding the projection of $\vec{b}$ onto $\vec{a}$. Here it is,

Vector Projection of $\vec{b}$ onto $\vec{a}$

$$\text{proj}_{\vec{a}}\vec{b} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|^2}\vec{a}$$

Note that we also need to be very careful with notation here. The projection of $\vec{a}$ onto $\vec{b}$ is given by

Vector Projection of $\vec{a}$ onto $\vec{b}$

$$\text{proj}_{\vec{b}}\,\vec{a} = \frac{\vec{a}\cdot\vec{b}}{\left\|\vec{b}\right\|^2}\vec{b}$$

We can see that this will be a totally different vector. This vector is parallel to $\vec{b}$, while $\text{proj}_{\vec{a}}\,\vec{b}$ is parallel to $\vec{a}$. So, be careful with notation and make sure you are finding the correct projection.

Here's an example.

Example 4

Determine the projection of $\vec{b} = \langle 2, 1, -1\rangle$ onto $\vec{a} = \langle 1, 0, -2\rangle$.

Solution

We need the dot product and the magnitude of $\vec{a}$.

$$\vec{a}\cdot\vec{b} = 4 \qquad \left\|\vec{a}\right\|^2 = 5$$

The projection is then,

$$\text{proj}_{\vec{a}}\,\vec{b} = \frac{\vec{a}\cdot\vec{b}}{\left\|\vec{a}\right\|^2}\vec{a} = \frac{4}{5}\langle 1, 0, -2\rangle = \left\langle \frac{4}{5}, 0, -\frac{8}{5}\right\rangle$$

For comparison purposes let's do it the other way around as well.

Example 5

Determine the projection of $\vec{a} = \langle 1, 0, -2\rangle$ onto $\vec{b} = \langle 2, 1, -1\rangle$.

Solution

We need the dot product and the magnitude of $\vec{b}$.

$$\vec{a}\cdot\vec{b} = 4 \qquad \left\|\vec{b}\right\|^2 = 6$$

The projection is then,

$$\text{proj}_{\vec{b}}\,\vec{a} = \frac{\vec{a}\cdot\vec{b}}{\left\|\vec{b}\right\|^2}\vec{b} = \frac{4}{6}\langle 2,1,-1\rangle = \left\langle \frac{4}{3},\frac{2}{3},-\frac{2}{3}\right\rangle$$

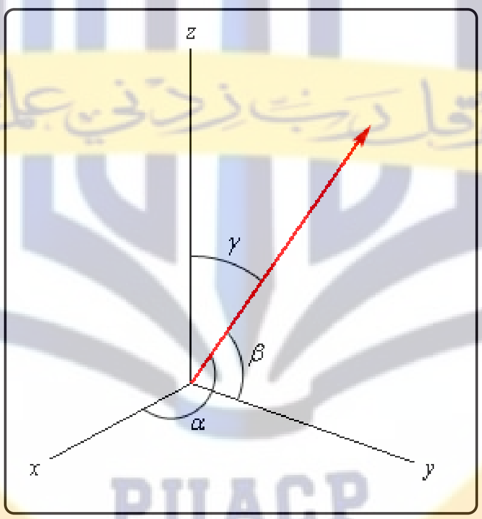
As we can see from the previous two examples the two projections are different so be careful.

Direction Cosines

This application of the dot product requires that we be in three dimensional space unlike all the other applications we've looked at to this point.

Let's start with a vector, $\vec{a}$, in three dimensional space. This vector will form angles with the x-axis (α), the y-axis (β), and the z-axis (γ). These angles are called **direction angles** and the cosines of these angles are called **direction cosines**.

Here is a sketch of a vector and the direction angles.



The formulas for the direction cosines are,

Direction Cosines

$$\cos(\alpha) = \frac{\vec{a}\cdot\vec{i}}{\|\vec{a}\|} = \frac{a_1}{\|\vec{a}\|} \qquad \cos(\beta) = \frac{\vec{a}\cdot\vec{j}}{\|\vec{a}\|} = \frac{a_2}{\|\vec{a}\|} \qquad \cos(\gamma) = \frac{\vec{a}\cdot\vec{k}}{\|\vec{a}\|} = \frac{a_3}{\|\vec{a}\|}$$

where $\vec{i}$, $\vec{j}$ and $\vec{k}$ are the standard basis vectors.

Let's verify the first dot product above. We'll leave the rest to you to verify.

$$\vec{a}\cdot\vec{i} = \langle a_1,a_2,a_3\rangle\cdot\langle 1,0,0\rangle = a_1$$

Here are a couple of nice facts about the direction cosines.

Facts

1. The vector $\vec{u} = \langle \cos(\alpha), \cos(\beta), \cos(\gamma) \rangle$ is a unit vector.
2. $\cos^2(\alpha) + \cos^2(\beta) + \cos^2(\gamma) = 1$
3. $\vec{a} = \|\vec{a}\| \langle \cos(\alpha), \cos(\beta), \cos(\gamma) \rangle$

Let's do a quick example involving direction cosines.

Example 6

Determine the direction cosines and direction angles for $\vec{a} = \langle 2, 1, -4 \rangle$.

Solution

We will need the magnitude of the vector.

$$\|\vec{a}\| = \sqrt{4 + 1 + 16} = \sqrt{21}$$

The direction cosines and angles are then,

$$\begin{aligned} \cos(\alpha) &= \frac{2}{\sqrt{21}} & \alpha &= 1.119 \text{ radians} = 64.123 \text{ degrees} \\ \cos(\beta) &= \frac{1}{\sqrt{21}} & \beta &= 1.351 \text{ radians} = 77.396 \text{ degrees} \\ \cos(\gamma) &= \frac{-4}{\sqrt{21}} & \gamma &= 2.632 \text{ radians} = 150.794 \text{ degrees} \end{aligned}$$

11.4 Cross Product

In this final section of this chapter we will look at the cross product of two vectors. We should note that the cross product requires both of the vectors to be three dimensional vectors.

Also, before getting into how to compute these we should point out a major difference between dot products and cross products. The result of a dot product is a number and the result of a cross product is a vector! Be careful not to confuse the two.

So, let's start with the two vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ then the cross product is given by the formula,

Cross Product - Formula

$$\vec{a} \times \vec{b} = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle$$

This is not an easy formula to remember. There are two ways to derive this formula. Both of them use the fact that the cross product is really the determinant of a 3x3 matrix. If you don't know what that is don't worry about it. You don't need to know anything about matrices or determinants to use either of the methods. The notation for the determinant is as follows,

Cross Product - Determinant

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

The first row is the standard basis vectors and must appear in the order given here. The second row is the components of $\vec{a}$ and the third row is the components of $\vec{b}$. Now, let's take a look at the different methods for getting the formula.

The first method uses the Method of Cofactors. If you don't know the method of cofactors that is fine, the result is all that we need. Here is the formula.

Method of Cofactors

$$\vec{a} \times \vec{b} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \vec{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \vec{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \vec{k}$$

where,

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

This formula is not as difficult to remember as it might at first appear to be. First, the terms alternate in sign and notice that the 2x2 is missing the column below the standard basis vector that multiplies it as well as the row of standard basis vectors.

The second method is slightly easier; however, many textbooks don't cover this method as it will only work on 3x3 determinants. This method says to take the determinant as listed above and then copy the first two columns onto the end as shown below.

$$\vec{a} \times \vec{b} = \left| \begin{array}{ccc|cc} \vec{i} & \vec{j} & \vec{k} & \vec{i} & \vec{j} \\ a_1 & a_2 & a_3 & a_1 & a_2 \\ b_1 & b_2 & b_3 & b_1 & b_2 \end{array} \right.$$

We now have three diagonals that move from left to right and three diagonals that move from right to left. We multiply along each diagonal and add those that move from left to right and subtract those that move from right to left.

This is best seen in an example. We'll also use this example to illustrate a fact about cross products.

Example 1

If $\vec{a} = \langle 2, 1, -1 \rangle$ and $\vec{b} = \langle -3, 4, 1 \rangle$ compute each of the following.

(a) $\vec{a} \times \vec{b}$

(b) $\vec{b} \times \vec{a}$

Solution

(a) $\vec{a} \times \vec{b}$

Here is the computation for this one.

$$\begin{aligned} \vec{a} \times \vec{b} &= \left| \begin{array}{ccc|cc} \vec{i} & \vec{j} & \vec{k} & \vec{i} & \vec{j} \\ 2 & 1 & -1 & 2 & 1 \\ -3 & 4 & 1 & -3 & 4 \end{array} \right. \\ &= \vec{i}\,(1)\,(1) + \vec{j}\,(-1)\,(-3) + \vec{k}\,(2)\,(4) - \vec{j}\,(2)\,(1) - \vec{i}\,(-1)\,(4) - \vec{k}\,(1)\,(-3) \\ &= 5\vec{i} + \vec{j} + 11\vec{k} \end{aligned}$$

(b) $\vec{b} \times \vec{a}$

And here is the computation for this one.

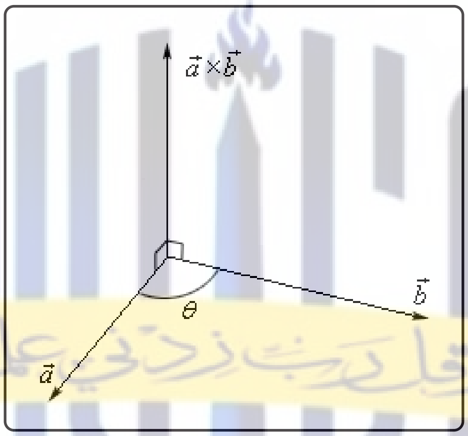
$$\begin{aligned} \vec{b} \times \vec{a} &= \left| \begin{array}{ccc|cc} \vec{i} & \vec{j} & \vec{k} & \vec{i} & \vec{j} \\ -3 & 4 & 1 & -3 & 4 \\ 2 & 1 & -1 & 2 & 1 \end{array} \right. \\ &= \vec{i}\,(4)\,(-1) + \vec{j}\,(1)\,(2) + \vec{k}\,(-3)\,(1) - \vec{j}\,(-3)\,(-1) - \vec{i}\,(1)\,(1) - \vec{k}\,(4)\,(2) \\ &= -5\vec{i} - \vec{j} - 11\vec{k} \end{aligned}$$

Notice that switching the order of the vectors in the cross product simply changed all the signs in the result. Note as well that this means that the two cross products will point in exactly opposite directions since they only differ by a sign. We'll formalize up this fact shortly when we list several facts.

There is also a geometric interpretation of the cross product. First we will let θ be the angle between the two vectors $\vec{a}$ and $\vec{b}$ and assume that $0 \le \theta \le \pi$, then we have the following fact,

$$\left\|\vec{a} \times \vec{b}\right\| = \|\vec{a}\| \ \left\|\vec{b}\right\| \sin(\theta) \tag{11.3}$$

and the following figure.



There should be a natural question at this point. How did we know that the cross product pointed in the direction that we've given it here?

First, as this figure implies, the cross product is orthogonal to both of the original vectors. This will always be the case with one exception that we'll get to in a second.

Second, we knew that it pointed in the upward direction (in this case) by the "right hand rule". This says that if we take our right hand, start at $\vec{a}$ and rotate our fingers towards $\vec{b}$ our thumb will point in the direction of the cross product. Therefore, if we'd sketched in $\vec{b} \times \vec{a}$ above we would have gotten a vector in the downward direction.

Example 2

A plane is defined by any three points that are in the plane. If a plane contains the points $P = (1, 0, 0)$, $Q = (1, 1, 1)$ and $R = (2, -1, 3)$ find a vector that is orthogonal to the plane.

Solution

The one way that we know to get an orthogonal vector is to take a cross product. So, if we could find two vectors that we knew were in the plane and took the cross product of these two vectors we know that the cross product would be orthogonal to both the vectors. However, since both the vectors are in the plane the cross product would then also be orthogonal to the plane.

So, we need two vectors that are in the plane. This is where the points come into the problem. Since all three points lie in the plane any vector between them must also be in the plane. There are many ways to get two vectors between these points. We will use the following two,

$$\overrightarrow{PQ} = \langle 1-1, 1-0, 1-0 \rangle = \langle 0, 1, 1 \rangle$$
$$\overrightarrow{PR} = \langle 2-1, -1-0, 3-0 \rangle = \langle 1, -1, 3 \rangle$$

The cross product of these two vectors will be orthogonal to the plane. So, let's find the cross product.

$$\begin{aligned} \overrightarrow{PQ} \times \overrightarrow{PR} &= \left| \begin{array}{ccc} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 1 \\ 1 & -1 & 3 \end{array} \right| \begin{array}{cc} \vec{i} & \vec{j} \\ 0 & 1 \\ 1 & -1 \end{array} \\ &= 4\vec{i} + \vec{j} - \vec{k} \end{aligned}$$

So, the vector $4\vec{i} + \vec{j} - \vec{k}$ will be orthogonal to the plane containing the three points.

Now, let's address the one time where the cross product will not be orthogonal to the original vectors. If the two vectors, $\vec{a}$ and $\vec{b}$, are parallel then the angle between them is either 0 or 180 degrees. From Equation 11.3 this implies that,

$$\left\| \vec{a} \times \vec{b} \right\| = 0$$

From a fact about the magnitude we saw in the first section we know that this implies

$$\vec{a} \times \vec{b} = \vec{0}$$

In other words, it won't be orthogonal to the original vectors since we have the zero vector. This does give us another test for parallel vectors however.

Fact

If $\vec{a} \times \vec{b} = \vec{0}$ then $\vec{a}$ and $\vec{b}$ will be parallel vectors.

Let's also formalize up the fact about the cross product being orthogonal to the original vectors.

Fact

Provided $\vec{a} \times \vec{b} \neq \vec{0}$ then $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$.

Here are some nice properties about the cross product.

Properties

If $\vec{u}$, $\vec{v}$ and $\vec{w}$ are vectors and c is a number then,

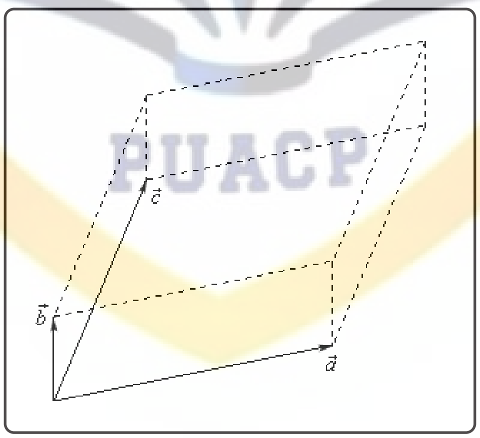
$$\vec{u} \times \vec{v} = -\vec{v} \times \vec{u} \qquad (c\vec{u}) \times \vec{v} = \vec{u} \times (c\vec{v}) = c\,(\vec{u} \times \vec{v})$$

$$\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w} \qquad \vec{u} \cdot (\vec{v} \times \vec{w}) = (\vec{u} \times \vec{v}) \cdot \vec{w}$$

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

The determinant in the last fact is computed in the same way that the cross product is computed. We will see an example of this computation shortly.

There are a couple of geometric applications to the cross product as well. Suppose we have three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ and we form the three dimensional figure shown below.



The area of the parallelogram (two dimensional front of this object) is given by,

$$\text{Area} = \left\| \vec{a} \times \vec{b} \right\|$$

and the volume of the parallelepiped (the whole three dimensional object) is given by,

$$\text{Volume} = \left| \vec{a} \cdot \left(\vec{b} \times \vec{c} \right) \right|$$

Note that the absolute value bars are required since the quantity could be negative and volume isn't negative.

We can use this volume fact to determine if three vectors lie in the same plane or not. If three vectors lie in the same plane then the volume of the parallelepiped will be zero.

Example 3

Determine if the three vectors $\vec{a} = \langle 1, 4, -7 \rangle$, $\vec{b} = \langle 2, -1, 4 \rangle$ and $\vec{c} = \langle 0, -9, 18 \rangle$ lie in the same plane or not.

Solution

So, as we noted prior to this example all we need to do is compute the volume of the parallelepiped formed by these three vectors. If the volume is zero they lie in the same plane and if the volume isn't zero they don't lie in the same plane.

$$\begin{aligned}
\vec{a} \cdot \left(\vec{b} \times \vec{c} \right) &= \left| \begin{array}{ccc} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{array} \right| \begin{array}{cc} 1 & 4 \\ 2 & -1 \\ 0 & -9 \end{array} \\
&= (1)(-1)(18) + (4)(4)(0) + (-7)(2)(-9) - \\
&\qquad (4)(2)(18) - (1)(4)(-9) - (-7)(-1)(0) \\
&= -18 + 126 - 144 + 36 \\
&= 0
\end{aligned}$$

So, the volume is zero and so they lie in the same plane.

12.6 Vector Functions

We first saw vector functions back when we were looking at the Equation of Lines. In that section we talked about them because we wrote down the equation of a line in $\mathbb{R}^3$ in terms of a *vector function* (sometimes called a *vector-valued function*). In this section we want to look a little closer at them and we also want to look at some vector functions in $\mathbb{R}^3$ other than lines.

A vector function is a function that takes one or more variables and returns a vector. We'll spend most of this section looking at vector functions of a single variable as most of the places where vector functions show up here will be vector functions of single variables. We will however briefly look at vector functions of two variables at the end of this section.

A vector functions of a single variable in $\mathbb{R}^2$ and $\mathbb{R}^3$ have the form,

$$\vec{r}(t) = \langle f(t), g(t) \rangle \qquad\qquad \vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

respectively, where $f(t)$, $g(t)$ and $h(t)$ are called the **component functions**.

The main idea that we want to discuss in this section is that of graphing and identifying the graph given by a vector function. Before we do that however, we should talk briefly about the domain of a vector function. The **domain** of a vector function is the set of all t's for which all the component functions are defined.

Example 1

Determine the domain of the following function.

$$\vec{r}(t) = \left\langle \cos(t), \ln(4-t), \sqrt{t+1} \right\rangle$$

Solution

The first component is defined for all t's. The second component is only defined for $t < 4$. The third component is only defined for $t \geq -1$. Putting all of these together gives the following domain.

$$[-1, 4)$$

This is the largest possible interval for which all three components are defined.

Let's now move into looking at the graph of vector functions. In order to graph a vector function all we do is think of the vector returned by the vector function as a position vector for points on the graph. Recall that a position vector, say $\vec{v} = \langle a, b, c \rangle$, is a vector that starts at the origin and ends at the point (a, b, c).

So, in order to sketch the graph of a vector function all we need to do is plug in some values of t and then plot points that correspond to the resulting position vector we get out of the vector function.

Because it is a little easier to visualize things we'll start off by looking at graphs of vector functions in $\mathbb{R}^2$.

Example 2

Sketch the graph of each of the following vector functions.

(a) $\vec{r}(t) = \langle t, 1 \rangle$

(b) $\vec{r}(t) = \langle t, t^3 - 10t + 7 \rangle$

Solution

(a) $\vec{r}(t) = \langle t, 1 \rangle$

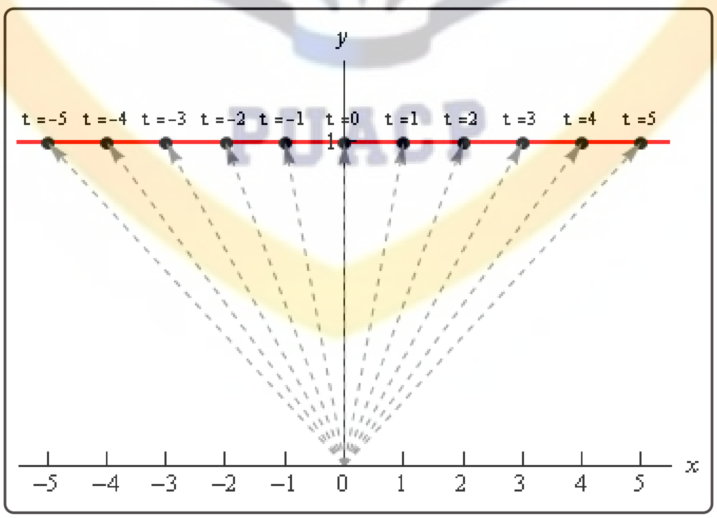
Okay, the first thing that we need to do is plug in a few values of t and get some position vectors. Here are a few,

$$\vec{r}(-3) = \langle -3, 1 \rangle \qquad \vec{r}(-1) = \langle -1, 1 \rangle \qquad \vec{r}(2) = \langle 2, 1 \rangle \qquad \vec{r}(5) = \langle 5, 1 \rangle$$

So, what this tells us is that the following points are all on the graph of this vector function.

$$(-3, 1) \qquad (-1, 1) \qquad (2, 1) \qquad (5, 1)$$

Here is a sketch of this vector function.



In this sketch we've included many more evaluations than just those above. Also note that we've put in the position vectors (in gray and dashed) so you can see how all this is working. Note however, that in practice the position vectors are generally not included in the sketch.

In this case it looks like we've got the graph of the line $y = 1$.

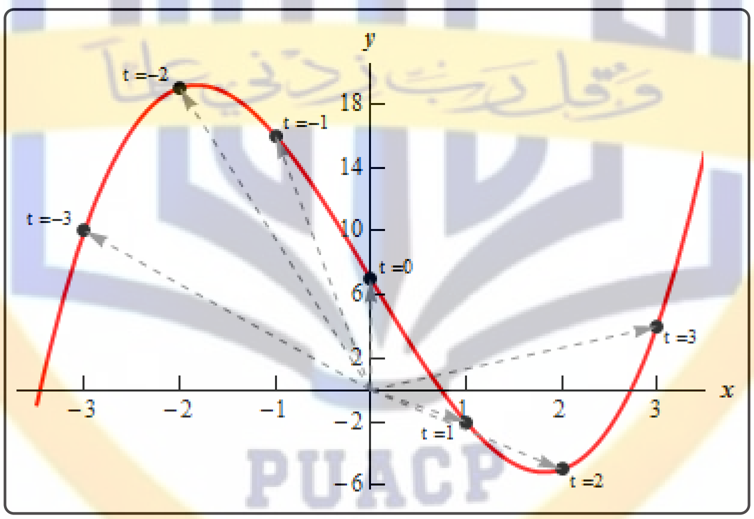
(b) $\vec{r}(t) = \langle t, t^3 - 10t + 7 \rangle$

Here are a couple of evaluations for this vector function.

$$\vec{r}(-3) = \langle -3, 10 \rangle \qquad \vec{r}(-1) = \langle -1, 16 \rangle \qquad \vec{r}(1) = \langle 1, -2 \rangle \qquad \vec{r}(3) = \langle 3, 4 \rangle$$

So, we've got a few points on the graph of this function. However, unlike the first part this isn't really going to be enough points to get a good idea of this graph. In general, it can take quite a few function evaluations to get an idea of what the graph is and it's usually easier to use a computer to do the graphing.

Here is a sketch of this graph. We've put in a few vectors/evaluations to illustrate them, but the reality is that we did have to use a computer to get a good sketch here.



Both of the vector functions in the above example were in the form,

$$\vec{r}(t) = \langle t, g(t) \rangle$$

and what we were really sketching is the graph of $y = g(x)$ as you probably caught onto. Let's graph a couple of other vector functions that do not fall into this pattern.

Example 3

Sketch the graph of each of the following vector functions.

(a) $\vec{r}(t) = \langle 6\cos(t), 3\sin(t)\rangle$

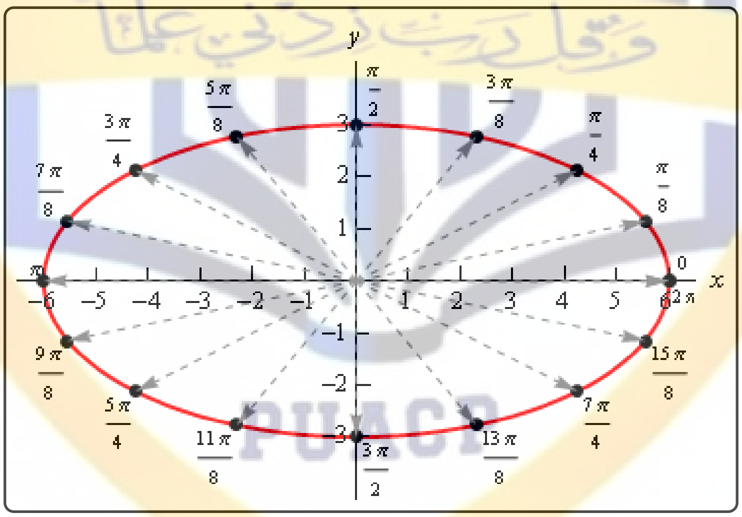
(b) $\vec{r}(t) = \langle t - 2\sin(t), t^2\rangle$

Solution

As we saw in the last part of the previous example it can really take quite a few function evaluations to really be able to sketch the graph of a vector function. Because of that we'll be skipping all the function evaluations here and just giving the graph. The main point behind this set of examples is to not get you too locked into the form we were looking at above. The first part will also lead to an important idea that we'll discuss after this example.

So, with that said here are the sketches of each of these.

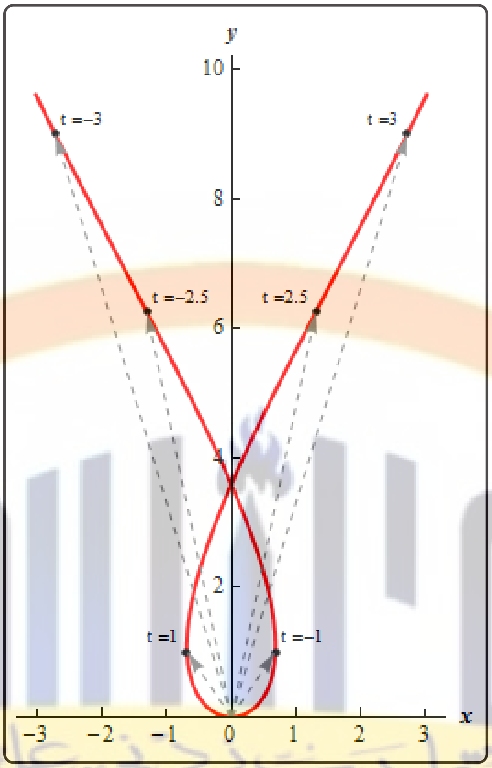
(a) $\vec{r}(t) = \langle 6\cos(t), 3\sin(t)\rangle$



So, in this case it looks like we've got an ellipse.

(b) $\vec{r}(t) = \langle t - 2\sin(t), t^2\rangle$

Here's the sketch for this vector function.



Before we move on to vector functions in $\mathbb{R}^3$ let's go back and take a quick look at the first vector function we sketched in the previous example, $\vec{r}(t) = \langle 6\cos(t), 3\sin(t) \rangle$. The fact that we got an ellipse here should not come as a surprise to you. We know that the first component function gives the x coordinate and the second component function gives the y coordinates of the point that we graph. If we strip these out to make this clear we get,

$$x = 6\cos(t) \qquad\qquad y = 3\sin(t)$$

This should look familiar to you. Back when we were looking at Parametric Equations we saw that this was nothing more than one of the sets of parametric equations that gave an ellipse.

This is an important idea in the study of vector functions. Any vector function can be broken down into a set of parametric equations that represent the same graph. In general, the two dimensional vector function, $\vec{r}(t) = \langle f(t), g(t) \rangle$, can be broken down into the parametric equations,

$$x = f(t) \qquad\qquad y = g(t)$$

Likewise, a three dimensional vector function, $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, can be broken down into the parametric equations,

$$x = f(t) \qquad y = g(t) \qquad z = h(t)$$

Do not get too excited about the fact that we're now looking at parametric equations in $\mathbb{R}^3$. They work in exactly the same manner as parametric equations in $\mathbb{R}^2$ which we're used to dealing with already. The only difference is that we now have a third component.

Let's take a look at a couple of graphs of vector functions.

Example 4

Sketch the graph of the following vector function.

$$\vec{r}(t) = \langle 2 - 4t, -1 + 5t, 3 + t \rangle$$

Solution

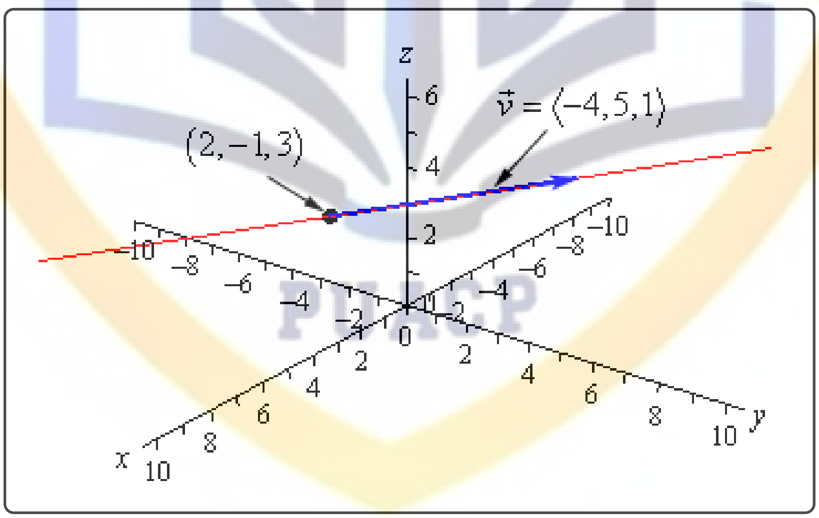
Notice that this is nothing more than a line. It might help if we rewrite it a little.

$$\vec{r}(t) = \langle 2, -1, 3 \rangle + t \langle -4, 5, 1 \rangle$$

In this form we can see that this is the equation of a line that goes through the point $(2, -1, 3)$ and is parallel to the vector $\vec{v} = \langle -4, 5, 1 \rangle$.

To graph this line all that we need to do is plot the point and then sketch in the parallel vector. In order to get the sketch will assume that the vector is on the line and will start at the point in the line. To sketch in the line all we do this is extend the parallel vector into a line.

Here is a sketch.



Example 5

Sketch the graph of the following vector function.

$$\vec{r}(t) = \langle 2\cos(t), 2\sin(t), 3\rangle$$

Solution

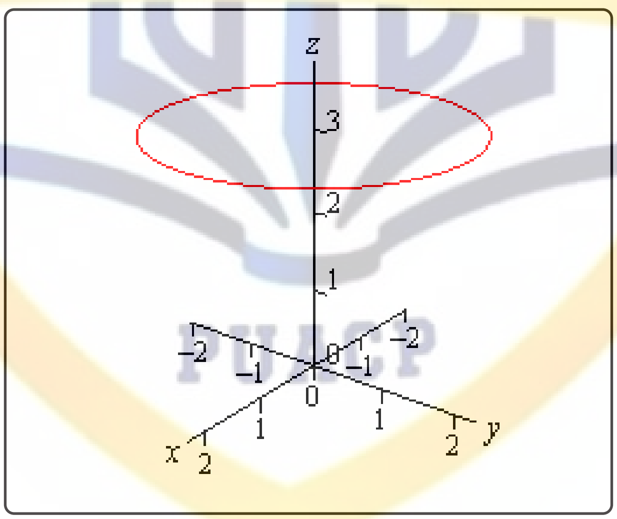
In this case to see what we've got for a graph let's get the parametric equations for the curve.

$$x = 2\cos(t) \qquad y = 2\sin(t) \qquad z = 3$$

If we ignore the z equation for a bit we'll recall (hopefully) that the parametric equations for x and y give a circle of radius 2 centered on the origin (or about the z-axis since we are in $\mathbb{R}^3$).

Now, all the parametric equations here tell us is that no matter what is going on in the graph all the z coordinates must be 3. So, we get a circle of radius 2 centered on the z-axis and at the level of $z = 3$.

Here is a sketch.



Note that it is very easy to modify the above vector function to get a circle centered on the x or y-axis as well. For instance,

$$\vec{r}(t) = \langle 10\sin(t), -3, 10\cos(t)\rangle$$

will be a circle of radius 10 centered on the y-axis and at $y = -3$. In other words, as long as two of the terms are a sine and a cosine (with the same coefficient) and the other is a fixed number then we will have a circle that is centered on the axis that is given by the fixed number.

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Let's take a look at a modification of this.

Example 6

Sketch the graph of the following vector function.

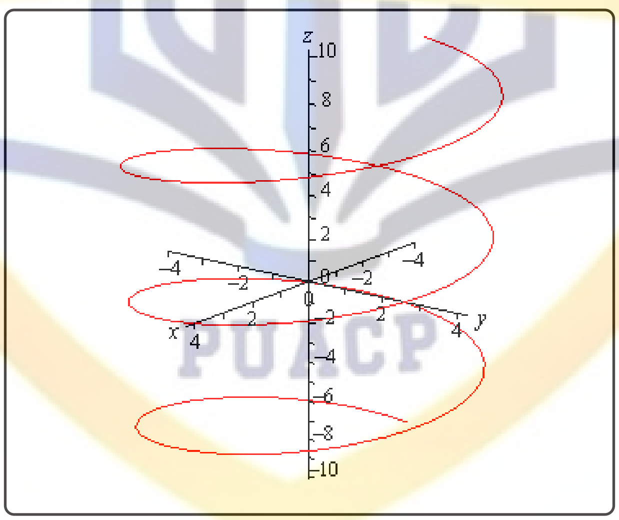
$$\vec{r}(t) = \langle 4\cos(t), 4\sin(t), t \rangle$$

Solution

If this one had a constant in the z component we would have another circle. However, in this case we don't have a constant. Instead we've got a t and that will change the curve. However, because the x and y component functions are still a circle in parametric equations our curve should have a circular nature to it in some way.

In fact, the only change is in the z component and as t increases the z coordinate will increase. Also, as t increases the x and y coordinates will continue to form a circle centered on the z-axis. Putting these two ideas together tells us that as we increase t the circle that is being traced out in the x and y directions should also be rising.

Here is a sketch of this curve.



So, we've got a helix (or spiral, depending on what you want to call it) here.

As with circles the component that has the t will determine the axis that the helix rotates about. For instance,

$$\vec{r}(t) = \langle t, 6\cos(t), 6\sin(t) \rangle$$

is a helix that rotates around the x-axis.

Also note that if we allow the coefficients on the sine and cosine for both the circle and helix to be different we will get ellipses.

For example,

$$\vec{r}(t) = \langle 9\cos(t), t, 2\sin(t)\rangle$$

will be a helix that rotates about the y-axis and is in the shape of an ellipse.

There is a nice formula that we should derive before moving onto vector functions of two variables.

Example 7

Determine the vector equation for the line segment starting at the point $P = (x_1, y_1, z_1)$ and ending at the point $Q = (x_2, y_2, z_2)$.

Solution

It is important to note here that we only want the equation of the line segment that starts at P and ends at Q. We don't want any other portion of the line and we do want the direction of the line segment preserved as we increase t. With all that said, let's not worry about that and just find the vector equation of the line that passes through the two points. Once we have this we will be able to get what we're after.

So, we need a point on the line. We've got two and we will use P. We need a vector that is parallel to the line and since we've got two points we can find the vector between them. This vector will lie on the line and hence be parallel to the line. Also, let's remember that we want to preserve the starting and ending point of the line segment so let's construct the vector using the same "orientation".

$$\vec{v} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1\rangle$$

Using this vector and the point P we get the following vector equation of the line.

$$\vec{r}(t) = \langle x_1, y_1, z_1\rangle + t\langle x_2 - x_1, y_2 - y_1, z_2 - z_1\rangle$$

While this is the vector equation of the line, let's rewrite the equation slightly.

$$\begin{aligned}\vec{r}(t) &= \langle x_1, y_1, z_1\rangle + t\langle x_2, y_2, z_2\rangle - t\langle x_1, y_1, z_1\rangle \\ &= (1-t)\langle x_1, y_1, z_1\rangle + t\langle x_2, y_2, z_2\rangle\end{aligned}$$

This is the equation of the line that contains the points P and Q. We of course just want the line segment that starts at P and ends at Q. We can get this by simply restricting the values of t.

Notice that

$$\vec{r}(0) = \langle x_1, y_1, z_1\rangle \qquad \vec{r}(1) = \langle x_2, y_2, z_2\rangle$$

So, if we restrict t to be between zero and one we will cover the line segment and we will start and end at the correct point.

So, the vector equation of the line segment that starts at $P = (x_1, y_1, z_1)$ and ends at $Q = (x_2, y_2, z_2)$ is,

$$\vec{r}(t) = (1-t)\langle x_1, y_1, z_1\rangle + t\langle x_2, y_2, z_2\rangle \qquad 0 \le t \le 1$$

As noted briefly at the beginning of this section we can also have vector functions of two variables. In these cases the graphs of vector function of two variables are surfaces. So, to make sure that we don't forget that let's work an example with that as well.

Example 8

Identify the surface that is described by $\vec{r}(x,y) = x\,\vec{i} + y\,\vec{j} + (x^2+y^2)\,\vec{k}$.

Solution

First, notice that in this case the vector function will in fact be a function of two variables. This will always be the case when we are using vector functions to represent surfaces.

To identify the surface let's go back to parametric equations.

$$x = x \qquad y = y \qquad z = x^2 + y^2$$

The first two are really only acknowledging that we are picking x and y for free and then determining z from our choices of these two. The last equation is the one that we want. We should recognize that function from the section on quadric surfaces. The third equation is the equation of an elliptic paraboloid and so the vector function represents an elliptic paraboloid.

As a final topic for this section let's generalize the idea from the previous example and note that given any function of one variable ($y = f(x)$ or $x = h(y)$) or any function of two variables ($z = g(x,y)$, $x = g(y,z)$, or $y = g(x,z)$) we can always write down a vector form of the equation.

For a function of one variable this will be,

$$\vec{r}(x) = x\,\vec{i} + f(x)\,\vec{j} \qquad \vec{r}(y) = h(y)\,\vec{i} + y\,\vec{j}$$

and for a function of two variables the vector form will be,

$$\vec{r}(x,y) = x\,\vec{i} + y\,\vec{j} + g(x,y)\,\vec{k} \qquad \vec{r}(y,z) = g(y,z)\,\vec{i} + y\,\vec{j} + z\,\vec{k}$$

$$\vec{r}(x,z) = x\,\vec{i} + g(x,z)\,\vec{j} + z\,\vec{k}$$

depending upon the original form of the function.

For example, the hyperbolic paraboloid $y = 2x^2 - 5z^2$ can be written as the following vector function.

$$\vec{r}(x,z) = x\,\vec{i} + \left(2x^2 - 5z^2\right)\,\vec{j} + z\,\vec{k}$$

This is a fairly important idea and we will be doing quite a bit of this kind of thing in multi-variable Calculus.

12.7 Calculus with Vector Functions

In this section we need to talk briefly about limits, derivatives and integrals of vector functions. As you will see, these behave in a fairly predictable manner. We will be doing all of the work in $\mathbb{R}^3$ but we can naturally extend the formulas/work in this section to $\mathbb{R}^n$ (*i.e.* n-dimensional space).

Let's start with limits. Here is the limit of a vector function.

Vector Function Limits

$$\begin{aligned}\lim_{t\to a} \vec{r}(t) &= \lim_{t\to a} \langle f(t), g(t), h(t)\rangle \\ &= \left\langle \lim_{t\to a} f(t), \lim_{t\to a} g(t), \lim_{t\to a} h(t)\right\rangle \\ &= \lim_{t\to a} f(t)\,\vec{i} + \lim_{t\to a} g(t)\,\vec{j} + \lim_{t\to a} h(t)\,\vec{k}\end{aligned}$$

So, all that we do is take the limit of each of the component's functions and leave it as a vector.

Example 1

Compute $\lim_{t\to 1} \vec{r}(t)$ where $\vec{r}(t) = \left\langle t^3, \dfrac{\sin(3t-3)}{t-1}, \mathbf{e}^{2t}\right\rangle$.

Solution

There really isn't all that much to do here.

$$\begin{aligned}\lim_{t\to 1} \vec{r}(t) &= \left\langle \lim_{t\to 1} t^3, \lim_{t\to 1} \frac{\sin(3t-3)}{t-1}, \lim_{t\to 1} \mathbf{e}^{2t}\right\rangle \\ &= \left\langle \lim_{t\to 1} t^3, \lim_{t\to 1} \frac{3\cos(3t-3)}{1}, \lim_{t\to 1} \mathbf{e}^{2t}\right\rangle \\ &= \left\langle 1, 3, \mathbf{e}^2\right\rangle\end{aligned}$$

Notice that we had to use L'Hospital's Rule on the y component.

Now let's take care of derivatives and after seeing how limits work it shouldn't be too surprising that we have the following for derivatives.

Vector Function Derivative

$$\vec{r}'(t) = \langle f'(t), g'(t), h'(t)\rangle = f'(t)\,\vec{i} + g'(t)\,\vec{j} + h'(t)\,\vec{k}$$

Example 2

Compute $\vec{r}'(t)$ for $\vec{r}(t) = t^6\,\vec{i} + \sin(2t)\,\vec{j} - \ln(t+1)\,\vec{k}$.

Solution

There really isn't too much to this problem other than taking the derivatives.

$$\vec{r}'(t) = 6t^5\,\vec{i} + 2\cos(2t)\,\vec{j} - \frac{1}{t+1}\vec{k}$$

Most of the basic facts that we know about derivatives still hold however, just to make it clear here are some facts about derivatives of vector functions.

Facts

$$\frac{d}{dt}\left(\vec{u} + \vec{v}\right) = \vec{u}\,' + \vec{v}\,'$$

$$\frac{d}{dt}\left(c\vec{u}\right) = c\,\vec{u}\,'$$

$$\frac{d}{dt}\left(f(t)\,\vec{u}(t)\right) = f'(t)\,\vec{u}(t) + f(t)\,\vec{u}\,'(t)$$

$$\frac{d}{dt}\left(\vec{u} \cdot \vec{v}\right) = \vec{u}\,' \cdot \vec{v} + \vec{u} \cdot \vec{v}\,'$$

$$\frac{d}{dt}\left(\vec{u} \times \vec{v}\right) = \vec{u}\,' \times \vec{v} + \vec{u} \times \vec{v}\,'$$

$$\frac{d}{dt}\left(\vec{u}(f(t))\right) = f'(t)\,\vec{u}\,'(f(t))$$

There is also one quick definition that we should get out of the way so that we can use it when we need to.

A **smooth curve** is any curve for which $\vec{r}'(t)$ is continuous and $\vec{r}'(t) \neq 0$ for any t except possibly at the endpoints. A helix is a smooth curve, for example.

Finally, we need to discuss integrals of vector functions. Using both limits and derivatives as a guide it shouldn't be too surprising that we also have the following for integration for indefinite integrals

Vector Function Indefinite Integral

$$\int \vec{r}(t)\,dt = \left\langle \int f(t)\,dt, \int g(t)\,dt, \int h(t)\,dt \right\rangle + \vec{c}$$
$$\int \vec{r}(t)\,dt = \int f(t)\,dt\,\vec{i} + \int g(t)\,dt\,\vec{j} + \int h(t)\,dt\,\vec{k} + \vec{c}$$

and the following for definite integrals.

Vector Function Definite Integral

$$\int_a^b \vec{r}(t)\,dt = \left\langle \int_a^b f(t)\,dt, \int_a^b g(t)\,dt, \int_a^b h(t)\,dt \right\rangle$$
$$\int_a^b \vec{r}(t)\,dt = \int_a^b f(t)\,dt\,\vec{i} + \int_a^b g(t)\,dt\,\vec{j} + \int_a^b h(t)\,dt\,\vec{k}$$

With the indefinite integrals we put in a constant of integration to make sure that it was clear that the constant in this case needs to be a vector instead of a regular constant.

Also, for the definite integrals we will sometimes write it as follows,

$$\int_a^b \vec{r}(t)\,dt = \left(\left\langle \int f(t)\,dt, \int g(t)\,dt, \int h(t)\,dt \right\rangle\right)\Bigg|_a^b$$
$$\int_a^b \vec{r}(t)\,dt = \left(\int f(t)\,dt\,\vec{i} + \int g(t)\,dt\,\vec{j} + \int h(t)\,dt\,\vec{k}\right)\Bigg|_a^b$$

In other words, we will do the indefinite integral and then do the evaluation of the vector as a whole instead of on a component by component basis.

Example 3

Compute $\int \vec{r}(t)\,dt$ for $\vec{r}(t) = \langle \sin(t), 6, 4t \rangle$.

Solution

All we need to do is integrate each of the components and be done with it.

$$\int \vec{r}(t)\,dt = \left\langle -\cos(t), 6t, 2t^2 \right\rangle + \vec{c}$$

Example 4

Compute $\int_0^1 \vec{r}(t)\, dt$ for $\vec{r}(t) = \langle \sin(t), 6, 4t \rangle$.

Solution

In this case all that we need to do is reuse the result from the previous example and then do the evaluation.

$$\begin{aligned}\int_0^1 \vec{r}(t)\, dt &= \left(\left\langle -\cos(t), 6t, 2t^2 \right\rangle \right) \Big|_0^1 \\ &= \langle -\cos(1), 6, 2 \rangle - \langle -1, 0, 0 \rangle \\ &= \langle 1 - \cos(1), 6, 2 \rangle\end{aligned}$$

12.8 Tangent, Normal and Binormal Vectors

In this section we want to look at an application of derivatives for vector functions. Actually, there are a couple of applications, but they all come back to needing the first one.

In the past we've used the fact that the derivative of a function was the slope of the tangent line. With vector functions we get exactly the same result, with one exception.

Given the vector function, $\vec{r}(t)$, we call $\vec{r}\,'(t)$ the **tangent vector** provided it exists and provided $\vec{r}\,'(t) \neq \vec{0}$. The tangent line to $\vec{r}(t)$ at P is then the line that passes through the point P and is parallel to the tangent vector, $\vec{r}\,'(t)$. Note that we really do need to require $\vec{r}\,'(t) \neq \vec{0}$ in order to have a tangent vector. If we had

$$\vec{r}\,'(t) = \vec{0}$$

we would have a vector that had no magnitude and so couldn't give us the direction of the tangent.

Also, provided $\vec{r}\,'(t) \neq \vec{0}$, the **unit tangent vector** to the curve is given by,

Unit Tangent Vector

$$\vec{T}(t) = \frac{\vec{r}\,'(t)}{\|\vec{r}\,'(t)\|}$$

While, the components of the unit tangent vector can be somewhat messy on occasion there are times when we will need to use the unit tangent vector instead of the tangent vector.

Example 1

Find the general formula for the tangent vector and unit tangent vector to the curve given by $\vec{r}(t) = t^2\,\vec{i} + 2\sin(t)\,\vec{j} + 2\cos(t)\,\vec{k}$.

Solution

First, by general formula we mean that we won't be plugging in a specific t and so we will be finding a formula that we can use at a later date if we'd like to find the tangent at any point on the curve. With that said there really isn't all that much to do at this point other than to do the work.

Here is the tangent vector to the curve.

$$\vec{r}\,'(t) = 2t\,\vec{i} + 2\cos(t)\,\vec{j} - 2\sin(t)\,\vec{k}$$

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To get the unit tangent vector we need the length of the tangent vector.

$$\begin{aligned}\left\|\vec{r}\,'(t)\right\| &= \sqrt{4t^2 + 4\cos^2(t) + 4\sin^2(t)}\\ &= \sqrt{4t^2+4}\end{aligned}$$

The unit tangent vector is then,

$$\begin{aligned}\vec{T}(t) &= \frac{1}{\sqrt{4t^2+4}}\left(2t\,\vec{i} + 2\cos t\,\vec{j} - 2\sin(t)\,\vec{k}\right)\\ &= \frac{2t}{\sqrt{4t^2+4}}\,\vec{i} + \frac{2\cos(t)}{\sqrt{4t^2+4}}\,\vec{j} - \frac{2\sin(t)}{\sqrt{4t^2+4}}\,\vec{k}\end{aligned}$$

Example 2

Find the vector equation of the tangent line to the curve given by $\vec{r}(t) = t^2\,\vec{i} + 2\sin(t)\,\vec{j} + 2\cos(t)\,\vec{k}$ at $t = \frac{\pi}{3}$.

Solution

First, we need the tangent vector and since this is the function we were working with in the previous example we can just reuse the tangent vector from that example and plug in $t = \frac{\pi}{3}$.

$$\vec{r}\,'\left(\frac{\pi}{3}\right) = \frac{2\pi}{3}\,\vec{i} + 2\cos\left(\frac{\pi}{3}\right)\,\vec{j} - 2\sin\left(\frac{\pi}{3}\right)\,\vec{k} = \frac{2\pi}{3}\,\vec{i} + \vec{j} - \sqrt{3}\,\vec{k}$$

We'll also need the point on the line at $t = \frac{\pi}{3}$ so,

$$\vec{r}\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}\,\vec{i} + \sqrt{3}\,\vec{j} + \vec{k}$$

The vector equation of the line is then,

$$\vec{r}(t) = \left\langle \frac{\pi^2}{9}, \sqrt{3}, 1\right\rangle + t\left\langle \frac{2\pi}{3}, 1, -\sqrt{3}\right\rangle$$

Before moving on let's note a couple of things about the previous example. First, we could have used the unit tangent vector had we wanted to for the parallel vector. However, that would have made for a more complicated equation for the tangent line.

Second, notice that we used $\vec{r}(t)$ to represent the tangent line despite the fact that we used that as well for the function. Do not get excited about that. The $\vec{r}(t)$ here is much like y is with normal functions. With normal functions, y is the generic letter that we used to represent functions and

$\vec{r}(t)$ tends to be used in the same way with vector functions.

Next, we need to talk about the **unit normal** and the **binormal** vectors.

The unit normal vector is defined to be,

Unit Normal Vector

$$\vec{N}(t) = \frac{\vec{T}'(t)}{\left\|\vec{T}'(t)\right\|}$$

The unit normal is orthogonal (or normal, or perpendicular) to the unit tangent vector and hence to the curve as well. We've already seen normal vectors when we were dealing with Equations of Planes. They will show up with some regularity in several Calculus III topics.

The definition of the unit normal vector always seems a little mysterious when you first see it. It follows directly from the following fact.

Fact

Suppose that $\vec{r}(t)$ is a vector such that $\|\vec{r}(t)\| = c$ for all t. Then $\vec{r}'(t)$ is orthogonal to $\vec{r}(t)$.

Proof

To prove this fact is pretty simple. From the fact statement and the relationship between the magnitude of a vector and the dot product we have the following.

$$\vec{r}(t) \cdot \vec{r}(t) = \|\vec{r}(t)\|^2 = c^2 \qquad \text{for all } t$$

Now, because this is true for all t we can see that,

$$\frac{d}{dt}\left(\vec{r}(t) \cdot \vec{r}(t)\right) = \frac{d}{dt}\left(c^2\right) = 0$$

Also, recalling the fact from the previous section about differentiating a dot product we see that,

$$\frac{d}{dt}\left(\vec{r}(t) \cdot \vec{r}(t)\right) = \vec{r}'(t) \cdot \vec{r}(t) + \vec{r}(t) \cdot \vec{r}'(t) = 2\vec{r}'(t) \cdot \vec{r}(t)$$

Or, upon putting all this together we get,

$$2\vec{r}'(t) \cdot \vec{r}(t) = 0 \qquad \Rightarrow \qquad \vec{r}'(t) \cdot \vec{r}(t) = 0$$

Therefore $\vec{r}'(t)$ is orthogonal to $\vec{r}(t)$.

The definition of the unit normal then falls directly from this. Because $\vec{T}(t)$ is a unit vector we know that $\left\|\vec{T}(t)\right\| = 1$ for all t and hence by the Fact $\vec{T}'(t)$ is orthogonal to $\vec{T}(t)$. However, because $\vec{T}(t)$ is tangent to the curve, $\vec{T}'(t)$ must be orthogonal, or normal, to the curve as well and so be a normal vector for the curve. All we need to do then is divide by $\left\|\vec{T}'(t)\right\|$ to arrive at a unit normal vector.

Next, is the binormal vector. The binormal vector is defined to be,

Binormal Vector

$$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$$

Because the binormal vector is defined to be the cross product of the unit tangent and unit normal vector we then know that the binormal vector is orthogonal to both the tangent vector and the normal vector.

Example 3

Find the normal and binormal vectors for $\vec{r}(t) = \langle t, 3\sin(t), 3\cos(t)\rangle$.

Solution

We first need the unit tangent vector so first get the tangent vector and its magnitude.

$$\vec{r}'(t) = \langle 1, 3\cos(t), -3\sin(t)\rangle$$
$$\left\|\vec{r}'(t)\right\| = \sqrt{1 + 9\cos^2(t) + 9\sin^2(t)} = \sqrt{10}$$

The unit tangent vector is then,

$$\vec{T}(t) = \left\langle \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\cos(t), -\frac{3}{\sqrt{10}}\sin(t)\right\rangle$$

The unit normal vector will now require the derivative of the unit tangent and its magnitude.

$$\vec{T}'(t) = \left\langle 0, -\frac{3}{\sqrt{10}}\sin(t), -\frac{3}{\sqrt{10}}\cos(t)\right\rangle$$
$$\left\|\vec{T}'(t)\right\| = \sqrt{\frac{9}{10}\sin^2(t) + \frac{9}{10}\cos^2(t)} = \sqrt{\frac{9}{10}} = \frac{3}{\sqrt{10}}$$

The unit normal vector is then,

$$\vec{N}(t) = \frac{\sqrt{10}}{3}\left\langle 0, -\frac{3}{\sqrt{10}}\sin(t), -\frac{3}{\sqrt{10}}\cos(t)\right\rangle = \langle 0, -\sin(t), -\cos(t)\rangle$$

Finally, the binormal vector is,

$$\begin{aligned}
\vec{B}(t) &= \vec{T}(t) \times \vec{N}(t) \\
&= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}}\cos(t) & -\frac{3}{\sqrt{10}}\sin(t) \\ 0 & -\sin(t) & -\cos(t) \end{vmatrix} \\
&= -\frac{3}{\sqrt{10}}\cos^2(t)\,\vec{i} - \frac{1}{\sqrt{10}}\sin(t)\,\vec{k} + \frac{1}{\sqrt{10}}\cos(t)\,\vec{j} - \frac{3}{\sqrt{10}}\sin^2(t)\,\vec{i} \\
&= -\frac{3}{\sqrt{10}}\,\vec{i} + \frac{1}{\sqrt{10}}\cos(t)\,\vec{j} - \frac{1}{\sqrt{10}}\sin(t)\,\vec{k}
\end{aligned}$$

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12.9 Arc Length with Vector Functions

In this section we'll recast an old formula into terms of vector functions. We want to determine the length of a vector function,

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

on the interval $a \le t \le b$.

We actually already know how to do this. Recall that we can write the vector function into the parametric form,

$$x = f(t) \qquad y = g(t) \qquad z = h(t)$$

Also, recall that with two dimensional parametric curves the arc length is given by,

$$L = \int_a^b \sqrt{\left[f'(t)\right]^2 + \left[g'(t)\right]^2}\, dt$$

There is a natural extension of this to three dimensions. So, the length of the curve $\vec{r}(t)$ on the interval $a \le t \le b$ is,

$$L = \int_a^b \sqrt{\left[f'(t)\right]^2 + \left[g'(t)\right]^2 + \left[h'(t)\right]^2}\, dt$$

There is a nice simplification that we can make for this. Notice that the integrand (the function we're integrating) is nothing more than the magnitude of the tangent vector,

$$\left\|\vec{r}\,'(t)\right\| = \sqrt{\left[f'(t)\right]^2 + \left[g'(t)\right]^2 + \left[h'(t)\right]^2}$$

Therefore, the arc length can be written as,

Arc Length

$$L = \int_a^b \left\|\vec{r}\,'(t)\right\| dt$$

Let's work a quick example of this.

Example 1

Determine the length of the curve $\vec{r}(t) = \langle 2t, 3\sin(2t), 3\cos(2t) \rangle$ on the interval $0 \le t \le 2\pi$.

Solution

We will first need the tangent vector and its magnitude.

$$\vec{r}'(t) = \langle 2, 6\cos(2t), -6\sin(2t)\rangle$$
$$\|\vec{r}'(t)\| = \sqrt{4 + 36\cos^2(2t) + 36\sin^2(2t)} = \sqrt{4+36} = 2\sqrt{10}$$

The length is then,

$$\begin{aligned} L &= \int_a^b \|\vec{r}'(t)\|\,dt \\ &= \int_0^{2\pi} 2\sqrt{10}\,dt \\ &= 4\pi\sqrt{10} \end{aligned}$$

We need to take a quick look at another concept here. We define the **arc length function** as,

Arc Length Function

$$s(t) = \int_0^t \|\vec{r}'(u)\|\,du$$

Before we look at why this might be important let's work a quick example.

Example 2

Determine the arc length function for $\vec{r}(t) = \langle 2t, 3\sin(2t), 3\cos(2t)\rangle$.

Solution

From the previous example we know that,

$$\|\vec{r}'(t)\| = 2\sqrt{10}$$

The arc length function is then,

$$s(t) = \int_0^t 2\sqrt{10}\,du = \left(2\sqrt{10}\,u\right)\Big|_0^t = 2\sqrt{10}\,t$$

Okay, just why would we want to do this? Well let's take the result of the example above and solve it for t.

$$t = \frac{s}{2\sqrt{10}}$$

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Now, taking this and plugging it into the original vector function and we can **reparametrize** the function into the form, $\vec{r}(t(s))$. For our function this is,

$$\vec{r}(t(s)) = \left\langle \frac{s}{\sqrt{10}}, 3\sin\left(\frac{s}{\sqrt{10}}\right), 3\cos\left(\frac{s}{\sqrt{10}}\right)\right\rangle$$

So, why would we want to do this? Well with the reparameterization we can now tell where we are on the curve after we've traveled a distance of s along the curve. Note as well that we will start the measurement of distance from where we are at $t = 0$.

Example 3

Where on the curve $\vec{r}(t) = \langle 2t, 3\sin(2t), 3\cos(2t)\rangle$ are we after traveling for a distance of $\dfrac{\pi\sqrt{10}}{3}$?

Solution

To determine this we need the reparameterization, which we have from above.

$$\vec{r}(t(s)) = \left\langle \frac{s}{\sqrt{10}}, 3\sin\left(\frac{s}{\sqrt{10}}\right), 3\cos\left(\frac{s}{\sqrt{10}}\right)\right\rangle$$

Then, to determine where we are all that we need to do is plug in $s = \frac{\pi\sqrt{10}}{3}$ into this and we'll get our location.

$$\vec{r}\left(t\left(\frac{\pi\sqrt{10}}{3}\right)\right) = \left\langle \frac{\pi}{3}, 3\sin\left(\frac{\pi}{3}\right), 3\cos\left(\frac{\pi}{3}\right)\right\rangle = \left\langle \frac{\pi}{3}, \frac{3\sqrt{3}}{2}, \frac{3}{2}\right\rangle$$

So, after traveling a distance of $\frac{\pi\sqrt{10}}{3}$ along the curve we are at the point $\left(\frac{\pi}{3}, \frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$.

12.10 Curvature

In this section we want to briefly discuss the **curvature** of a smooth curve (recall that for a smooth curve we require $\vec{r}'(t)$ is continuous and $\vec{r}'(t) \neq 0$). The curvature measures how fast a curve is changing direction at a given point.

There are several formulas for determining the curvature for a curve. The formal definition of curvature is,

$$\kappa = \left\| \frac{d\vec{T}}{ds} \right\|$$

where $\vec{T}$ is the unit tangent and s is the arc length. Recall that we saw in a previous section how to reparametrize a curve to get it into terms of the arc length.

In general the formal definition of the curvature is not easy to use so there are two alternate formulas that we can use. Here they are.

Curvature

$$\kappa = \frac{\left\|\vec{T}'(t)\right\|}{\left\|\vec{r}'(t)\right\|} \qquad\qquad \kappa = \frac{\left\|\vec{r}'(t) \times \vec{r}''(t)\right\|}{\left\|\vec{r}'(t)\right\|^3}$$

These may not be particularly easy to deal with either, but at least we don't need to reparametrize the unit tangent.

Example 1

Determine the curvature for $\vec{r}(t) = \langle t, 3\sin(t), 3\cos(t)\rangle$.

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Solution

Back in the section when we introduced the tangent vector we computed the tangent and unit tangent vectors for this function. These were,

$$\vec{r}'(t) = \langle 1, 3\cos(t), -3\sin(t)\rangle$$
$$\vec{T}(t) = \left\langle \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\cos(t), -\frac{3}{\sqrt{10}}\sin(t)\right\rangle$$

The derivative of the unit tangent is,

$$\vec{T}'(t) = \left\langle 0, -\frac{3}{\sqrt{10}}\sin(t), -\frac{3}{\sqrt{10}}\cos(t)\right\rangle$$

The magnitudes of the two vectors are,

$$\left\|\vec{r}'(t)\right\| = \sqrt{1 + 9\cos^2(t) + 9\sin^2(t)} = \sqrt{10}$$
$$\left\|\vec{T}'(t)\right\| = \sqrt{0 + \frac{9}{10}\sin^2(t) + \frac{9}{10}\cos^2(t)} = \sqrt{\frac{9}{10}} = \frac{3}{\sqrt{10}}$$

The curvature is then,

$$\kappa = \frac{\left\|\vec{T}'(t)\right\|}{\left\|\vec{r}'(t)\right\|} = \frac{{}^{3}/_{\sqrt{10}}}{\sqrt{10}} = \frac{3}{10}$$

In this case the curvature is constant. This means that the curve is changing direction at the same rate at every point along it. Recalling that this curve is a helix this result makes sense.

Example 2

Determine the curvature of $\vec{r}(t) = t^2\,\vec{i} + t\,\vec{k}$.

Solution

In this case the second form of the curvature would probably be easiest. Here are the first couple of derivatives.

$$\vec{r}'(t) = 2t\,\vec{i} + \vec{k} \qquad\qquad \vec{r}''(t) = 2\,\vec{i}$$

Next, we need the cross product.

$$\vec{r}'(t) \times \vec{r}''(t) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2t & 0 & 1 \\ 2 & 0 & 0 \end{vmatrix}$$
$$= 2\vec{j}$$

The magnitudes are,

$$\left\|\vec{r}'(t) \times \vec{r}''(t)\right\| = 2 \qquad\qquad \left\|\vec{r}'(t)\right\| = \sqrt{4t^2 + 1}$$

The curvature at any value of t is then,

$$\kappa = \frac{2}{\left(4t^2 + 1\right)^{\frac{3}{2}}}$$

There is a special case that we can look at here as well. Suppose that we have a curve given by

$y = f(x)$ and we want to find its curvature.

As we saw when we first looked at vector functions we can write this as follows,

$$\vec{r}(x) = x\,\vec{i} + f(x)\,\vec{j}$$

If we then use the second formula for the curvature we will arrive at the following formula for the curvature.

$$\kappa = \frac{|f''(x)|}{\left(1 + [f'(x)]^2\right)^{\frac{3}{2}}}$$

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12.11 Velocity and Acceleration

In this section we need to take a look at the velocity and acceleration of a moving object.

From Calculus I we know that given the position function of an object that the velocity of the object is the first derivative of the position function and the acceleration of the object is the second derivative of the position function.

So, given this it shouldn't be too surprising that if the position function of an object is given by the vector function $\vec{r}(t)$ then the velocity and acceleration of the object is given by,

$$\vec{v}(t) = \vec{r}\,'(t) \qquad\qquad \vec{a}(t) = \vec{r}\,''(t)$$

Notice that the velocity and acceleration are also going to be vectors as well.

In the study of the motion of objects the acceleration is often broken up into a **tangential component**, a_T, and a **normal component**, a_N. The tangential component is the part of the acceleration that is tangential to the curve and the normal component is the part of the acceleration that is normal (or orthogonal) to the curve. If we do this we can write the acceleration as,

$$\vec{a} = a_T \vec{T} + a_N \vec{N}$$

where $\vec{T}$ and $\vec{N}$ are the unit tangent and unit normal for the position function.

If we define $v = \|\vec{v}(t)\|$ then the tangential and normal components of the acceleration are given by,

Tangential and Normal Acceleration

$$a_T = v' = \frac{\vec{r}\,'(t) \cdot \vec{r}\,''(t)}{\|\vec{r}\,'(t)\|} \qquad\qquad a_N = \kappa v^2 = \frac{\|\vec{r}\,'(t) \times \vec{r}\,''(t)\|}{\|\vec{r}\,'(t)\|}$$

where κ is the curvature for the position function.

There are two formulas to use here for each component of the acceleration and while the second formula may seem overly complicated it is often the easier of the two. In the tangential component, v, may be messy and computing the derivative may be unpleasant. In the normal component we will already be computing both of these quantities in order to get the curvature and so the second formula in this case is definitely the easier of the two.

Let's take a quick look at a couple of examples.

Example 1

If the acceleration of an object is given by $\vec{a} = \vec{i} + 2\vec{j} + 6t\vec{k}$ find the object's velocity and position functions given that the initial velocity is $\vec{v}(0) = \vec{j} - \vec{k}$ and the initial position is $\vec{r}(0) = \vec{i} - 2\vec{j} + 3\vec{k}$.

Solution

We'll first get the velocity. To do this all (well almost all) we need to do is integrate the acceleration.

$$\begin{aligned}\vec{v}(t) &= \int \vec{a}(t)\,dt \\ &= \int \vec{i} + 2\vec{j} + 6t\vec{k}\,dt \\ &= t\,\vec{i} + 2t\,\vec{j} + 3t^2\,\vec{k} + \vec{c}\end{aligned}$$

To completely get the velocity we will need to determine the "constant" of integration. We can use the initial velocity to get this.

$$\vec{j} - \vec{k} = \vec{v}(0) = \vec{c}$$

The velocity of the object is then,

$$\begin{aligned}\vec{v}(t) &= t\,\vec{i} + 2t\,\vec{j} + 3t^2\,\vec{k} + \vec{j} - \vec{k} \\ &= t\,\vec{i} + (2t+1)\,\vec{j} + \left(3t^2 - 1\right)\vec{k}\end{aligned}$$

We will find the position function by integrating the velocity function.

$$\begin{aligned}\vec{r}(t) &= \int \vec{v}(t)\,dt \\ &= \int t\,\vec{i} + (2t+1)\,\vec{j} + \left(3t^2 - 1\right)\vec{k}\,dt \\ &= \frac{1}{2}t^2\,\vec{i} + \left(t^2 + t\right)\vec{j} + \left(t^3 - t\right)\vec{k} + \vec{c}\end{aligned}$$

Using the initial position gives us,

$$\vec{i} - 2\vec{j} + 3\vec{k} = \vec{r}(0) = \vec{c}$$

So, the position function is,

$$\vec{r}(t) = \left(\frac{1}{2}t^2 + 1\right)\vec{i} + \left(t^2 + t - 2\right)\vec{j} + \left(t^3 - t + 3\right)\vec{k}$$

Example 2

For the object in the previous example determine the tangential and normal components of the acceleration.

Solution

There really isn't much to do here other than plug into the formulas. To do this we'll need to notice that,

$$\begin{aligned}\vec{r}'(t) &= t\,\vec{i} + (2t+1)\,\vec{j} + \left(3t^2-1\right)\vec{k}\\ \vec{r}''(t) &= \vec{i} + 2\vec{j} + 6t\vec{k}\end{aligned}$$

Let's first compute the dot product and cross product that we'll need for the formulas.

$$\vec{r}'(t)\cdot\vec{r}''(t) = t + 2(2t+1) + 6t\left(3t^2-1\right) = 18t^3 - t + 2$$

$$\begin{aligned}\vec{r}'(t)\times\vec{r}''(t) &= \begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ t & 2t+1 & 3t^2-1\\ 1 & 2 & 6t\end{vmatrix}\\ &= (6t)(2t+1)\,\vec{i} + \left(3t^2-1\right)\vec{j} + 2t\vec{k} - 6t^2\vec{j} - 2\left(3t^2-1\right)\vec{i} - (2t+1)\,\vec{k}\\ &= \left(6t^2+6t+2\right)\vec{i} - \left(3t^2+1\right)\vec{j} - \vec{k}\end{aligned}$$

Next, we also need a couple of magnitudes.

$$\begin{aligned}\left\|\vec{r}'(t)\right\| &= \sqrt{t^2 + (2t+1)^2 + \left(3t^2-1\right)^2} = \sqrt{9t^4 - t^2 + 4t + 2}\\ \left\|\vec{r}'(t)\times\vec{r}''(t)\right\| &= \sqrt{\left(6t^2+6t+2\right)^2 + \left(3t^2+1\right)^2 + 1} = \sqrt{45t^4 + 72t^3 + 66t^2 + 24t + 6}\end{aligned}$$

The tangential component of the acceleration is then,

$$a_T = \frac{18t^3 - t + 2}{\sqrt{9t^4 - t^2 + 4t + 2}}$$

The normal component of the acceleration is,

$$a_N = \frac{\sqrt{45t^4 + 72t^3 + 66t^2 + 24t + 6}}{\sqrt{9t^4 - t^2 + 4t + 2}} = \sqrt{\frac{45t^4 + 72t^3 + 66t^2 + 24t + 6}{9t^4 - t^2 + 4t + 2}}$$